



Efficient parallel computing with a compact finite difference scheme

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ABSTRACT

This paper proposes an efficient parallel computing approach based on a high-order accurate compact finite difference scheme in conjunction with a conventional domain decomposition method and MPI libraries. The proposed parallel computing approach consists of two major features: (a) a newly developed compact finite difference scheme with extended stencils containing halo points around subdomain boundaries, and (b) a predictor–corrector type implementation of a compact filter that effectively suppresses spurious errors from the subdomain boundaries. The current work employs three halo cells for the inter-node communication, based on which the coefficients of the new compact scheme at the subdomain boundaries are optimized to achieve as high level of resolution and accuracy as the interior compact scheme provides. Also, an optimal set of cut-off wavenumbers of the compact filter that minimizes spurious errors is suggested. It is shown that the level of errors from the proposed parallel calculations lies within the same order of magnitude of that from the single-domain serial calculations. The overall accuracy and linear stability of the new parallel compact differencing–filtering system are confirmed by grid convergence tests and eigenvalue analyses. The proposed approach shows a substantial improvement with respect to existing methods available.

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1. Introduction

Over the last decade, compact finite difference schemes have become widespread in direct numerical simulation (DNS), large eddy simulation (LES) and computational aeroacoustics (CAA) because they provide both high-order accuracy and high-resolution characteristics via optimization of the coefficients [1–5]. Currently, almost every research activity in the area of CAA/LES/DNS makes extensive use of parallel computing techniques owing to the rapid growth of modern supercomputing capabilities. However, the use of the compact finite difference schemes in a parallel computing environment is not straightforward, particularly when domain decomposition with MPI (message passing interface) libraries is concerned. This is mainly due to their implicit nature associated with the solution of banded Hermitian matrix systems. Common approaches for parallelizing such systems include: reduced parallel diagonal dominant algorithm [6]; alternating-direction line-relaxation method [7]; pipelined implementation of Thomas algorithm [8]; and using a large area of overlap between two neighboring subdomains [9]. These approaches, currently available for tridiagonal matrix systems, are significantly more expensive in terms of inter-node communication than parallelizing standard finite

difference schemes, which undermines the genuine advantage of using compact schemes.

Recently, Sengupta et al. [9], after reviewing the current state-of-the-art of parallel computing strategies for compact schemes, suggested that the implementation of compact schemes in an overlapped fashion with inclusion of filters might be used as an efficient alternative to earlier more computationally expensive parallelization strategies. They developed a new tridiagonal compact scheme for this purpose incorporating six-point overlap between two adjacent subdomains. However, their benchmark tests on vortex convection showed a controversial behavior of their scheme resulting in higher error on a finer grid [9]. Also, some of the unaddressed issues in their approach include: whether the parallelized system achieves the desired grid convergence rate, and how much improvement could be made with different filters or cut-off wavenumbers. These are important questions that need to be answered. This paper takes all these issues into account and aims to deliver a more reliable and efficient strategy of parallelizing compact schemes based on a halo-cell approach instead of the overlapping method that demands more operations and memory.

In the current work the main platform is based on a pentadiagonal compact scheme contrary to the other above mentioned works that are currently limited to tridiagonal schemes. The new platform employs one of the latest pentadiagonal compact schemes with fourth-order accuracy proposed by Kim [5] and extends his boundary formulations to accommodate wider stencils that contain three halo cells from the adjacent subdomains.

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Although the new compact scheme across the subdomain boundaries does not precisely reproduce the interior resolution, the loss of accuracy can be minimized. The new platform also includes a pentadiagonal compact filter with sixth-order accuracy [10] to remove spurious errors, and introduces a predictor–corrector type of execution in order to achieve almost identical error levels to those from single-domain serial calculations. The compact filter has a variable cut-off wavenumber that may change locally across the subdomain boundaries, and this helps improve the filter performance. Eigenvalue analyses and grid convergence tests confirm that the entire differencing–filtering system provides linear stability and fourth-order accuracy throughout the domain. The proposed parallel computing approach is rigorously tested through a variety of one-, two- and three-dimensional test cases using fully nonlinear compressible Euler and Navier–Stokes equations in both Cartesian and generalized coordinates.

The paper is organized as follows. Section 2 shows the development of the new compact finite difference scheme with halo points. Section 3 describes the implementation of a compact filter based on predictor–corrector steps. Section 4 performs grid convergence tests and linear stability analysis of the proposed compact differencing–filtering system. Section 5 briefly introduces the governing equations solved for the numerical tests. The results of Euler and direct numerical simulations are presented and discussed in Sections 6 and 7. Performance and efficiency of the proposed parallel computing approach is demonstrated in Section 8. Finally concluding remarks are made in Section 9.

2. New compact finite difference scheme with halo points

This section presents a new compact finite difference scheme developed for parallel computing based on domain decomposition and MPI (message passing interface) libraries. It consists of a central interior scheme and non-central boundary schemes that involve three halo cells from the adjacent subdomain. Fourth-order accuracy is maintained throughout the domain. The new boundary schemes are specifically designed to construct a closed pentadiagonal matrix system within each subdomain. They are optimized to achieve similar resolution characteristics to those of the interior scheme.

2.1. Interior compact finite difference scheme

The present work employs a fourth-order central compact scheme [5] for interior points, which is based on a pentadiagonal platform and a seven-point stencil. It may be expressed as

$$\beta \bar{f}'_{i-2} + \alpha \bar{f}'_{i-1} + \bar{f}'_i + \alpha \bar{f}'_{i+1} + \beta \bar{f}'_{i+2} = \frac{1}{\Delta x} \sum_{m=1}^3 a_m (f_{i+m} - f_{i-m}) \quad \text{for } 2 \leq i \leq N-2 \quad (2.1)$$

where f_i and f'_i represent an objective function $f(x)$ and its spatial derivative $\partial f(x)/\partial x$ respectively at a location of interest x_i . The bar “–” is used in order to distinguish numerical derivative ($\bar{f}'$) from the exact derivative (f'). The spatial interval $\Delta x = x_{i+1} - x_i$ is a constant independent of the index i in the computational domain where all the grid points are equally spaced. The coefficients α , β , a_1 , a_2 and a_3 optimized in [5] are listed in Table 1. The index runs

through $0 \leq i \leq N$ within a subdomain, where $i = 0$ and $i = N$ represent the subdomain boundaries. Accordingly, $i \leq -1$ or $i \geq N+1$ indicate halo points from the adjacent subdomains. Eq. (2.1) being used at $i = 2$ or $i = N-2$ involves one of the halo points (f_{-1} or f_{N+1}) on the right-hand-side.

With Eq. (2.1) used for the interior points ($i \geq 2$), two additional schemes are required for the boundary points ($i = 0$ and 1) in order to close the pentadiagonal matrix and enable independent inversion of the matrix in each subdomain. This approach demands less inter-node communication than the earlier approaches mentioned in Section 1. The formulation of the two additional schemes obtained by using an extrapolation technique and optimization procedure is described in the following.

2.2. Formulation of boundary schemes with halo points

It is obvious that Eq. (2.1) applies directly on the interior points. To be able to keep applying it at the boundary points ($i = 0$ and 1), an extrapolation may be used to approximate the unknown derivatives $f'(x)$ in the halo area and substitute them with interior (or known) terms. The following is a spline function from the boundary ($x = x_0$) and its first derivative that may be used for the extrapolation [5]:

$$g(x^*) = \sum_{m=0}^{N_A} p_m x^{*m} + \sum_{m=1}^{N_B} [q_m \cos(\phi_m x^*) + r_m \sin(\phi_m x^*)] \quad (2.2)$$

$$g'(x^*) = \frac{dg(x^*)}{dx} = \frac{1}{\Delta x} \left\{ \sum_{m=1}^{N_A} m p_m x^{*m-1} - \sum_{m=1}^{N_B} \phi_m [q_m \sin(\phi_m x^*) - r_m \cos(\phi_m x^*)] \right\} \quad (2.3)$$

where $x^* = (x - x_0)/\Delta x$ is the non-dimensional coordinate from the boundary. The extrapolation function is a linear combination of polynomials and trigonometric series. The constants N_A and N_B represent the orders of each series. In this paper, $N_A = 4$ is selected to maintain the fourth-order accuracy and $N_B = 3$ to match the number of coefficients and available constraints. The coefficients p_m ($m = 0, \dots, N_A$), q_m and r_m ($m = 1, \dots, N_B$) should be determined by matching constraints as described below. The control variables ϕ_m ($m = 1, \dots, N_B$) are introduced to optimize the resulting schemes for the best resolution characteristics.

Eleven matching constraints are necessary since the number of coefficients to be determined is $1 + N_A + 2N_B = 11$, and they are as follows:

$$g(m) = f_m \quad \text{for } m = -3, -2, -1 (\text{halo points}) \text{ and } 0, \dots, 4, \quad (2.4)$$

$$g'(m) = \bar{f}'_m \quad \text{for } m = 0, 1, 2. \quad (2.5)$$

Solving Eqs. (2.4) and (2.5) leads to the coefficients p_m ($m = 0, \dots, 4$), q_m and r_m ($m = 1, 2, 3$) being replaced by linear combinations of the known function values ($f_{-3}, \dots, f_4$) and the interior derivatives ($\bar{f}'_0, \bar{f}'_1, \bar{f}'_2$). However, the final coefficients are yet to be determined once the control variables ϕ_1 , ϕ_2 and ϕ_3 have been fixed, which is shown later. Once all the coefficients are determined,

Table 1
Coefficients for Eq. (2.1).

α	β	a_1	a_2	a_3
0.5862704032801503	0.9549533555017055e–1	0.6431406736919156	0.2586011023495066	0.7140953479797375e–2

it is possible to apply Eq. (2.1) at the boundary points ($i = 0$ and 1) as follows:

$$\begin{aligned} i = 0 : \quad & \beta g'(-2) + \alpha g'(-1) + \bar{f}_0' + \alpha \bar{f}_1' + \beta \bar{f}_2' \\ & = \frac{1}{\Delta x} \sum_{m=1}^3 a_m (f_m - f_{-m}), \end{aligned} \quad (2.6)$$

$$\begin{aligned} i = 1 : \quad & \beta g'(-1) + \alpha \bar{f}_0' + \bar{f}_1' + \alpha \bar{f}_2' + \beta \bar{f}_3' \\ & = \frac{1}{\Delta x} \sum_{m=1}^3 a_m (f_{1+m} - f_{1-m}). \end{aligned} \quad (2.7)$$

Implementing the extrapolation for $g'(-1)$ and $g'(-2)$ using Eq. (2.3) may turn Eqs. (2.6) and (2.7) into the following form:

$$\sum_{m=0}^M c_m \bar{f}_m' = \frac{1}{\Delta x} \sum_{m=-3}^4 d_m f_m \quad \text{for } i = 0, 1 \quad (2.8)$$

where $M = 2$ and 3 for the cases of $i = 0$ and 1 , respectively. The intermediary coefficients c_m ($m = 0, \dots, M$) and d_m ($m = -3, \dots, 4$) are given by certain linear combinations of p_m ($m = 0, \dots, 4$), q_m and r_m ($m = 1, 2, 3$) as a result of the constraints in Eqs. (2.4) and (2.5).

Dividing Eq. (2.8) with c_n ($n = 0$ and 1 for the cases of $i = 0$ and 1 , respectively) for normalization and rearranging the right-hand-side terms ultimately results in the following equations:

$$i = 0 : \quad \bar{f}_0' + \gamma_{01} \bar{f}_1' + \gamma_{02} \bar{f}_2' = \frac{1}{\Delta x} \sum_{m=-3, \neq 0}^4 b_{0m} (f_m - f_0), \quad (2.9)$$

$$\begin{aligned} i = 1 : \quad & \gamma_{10} \bar{f}_0' + \bar{f}_1' + \gamma_{12} \bar{f}_2' + \gamma_{13} \bar{f}_3' \\ & = \frac{1}{\Delta x} \sum_{m=-3, \neq 1}^4 b_{1m} (f_m - f_1). \end{aligned} \quad (2.10)$$

These are fourth-order compact finite difference schemes with eight-point stencils including three halo points from the adjacent subdomain. This formulation enables to close the pentadiagonal matrix system and solve it within each subdomain once halo data have been exchanged. The next task is to find an optimal set of the control variables (ϕ_1, ϕ_2, ϕ_3) in order to maximize the resolution of the schemes. The optimization task is of critical importance since Eqs. (2.9) and (2.10) are non-central schemes that might cause substantial (mainly dispersive) errors across the subdomain boundaries if the coefficients are not carefully determined.

2.3. Optimization of boundary schemes via Fourier analysis

Dispersive errors from the non-central schemes at subdomain boundaries are one of the major obstacles in the current halo-cell based parallel computing approach. It is crucial that each subdomain boundary must not generate dispersive errors that could eventually mask the entire acoustic field when aerodynamic sound is concerned. An optimization of the schemes is presented here in order to reduce the dispersive errors to a level sufficiently low to allow for precise aeroacoustic calculations in massively parallel computing environments. The optimization is implemented in the wavenumber domain via Fourier transform. Mathematically, the Fourier transform of a continuous function (subject to certain conditions – not to be discussed in detail here) is defined as

$$\tilde{f}(k) = \int_{-\infty}^{\infty} f(x) e^{-jkx} dx \quad (2.11)$$

where $j = \sqrt{-1}$, k is the wavenumber and the tilde represents the transformed function. The use of Fourier transform to analyze difference approximations has been discussed and established through a series of publications for the last five decades as listed in [11].

With f_{i+m} and $\bar{f}_{i+m}'$ replaced by $f(x + m\Delta x)$ and $\bar{f}'(x + m\Delta x)$, taking the Fourier transform of Eqs. (2.9) and (2.10) results in the following equation:

$$j\bar{\kappa}_i [A_i(\kappa) + jB_i(\kappa)] = C_i(\kappa) + jD_i(\kappa) \quad \text{for } i = 0, 1 \quad (2.12)$$

where $\kappa = k\Delta x$ is a scaled wavenumber and $\bar{\kappa} = \bar{k}\Delta x$ is a scaled pseudo-wavenumber which deviates from the true wavenumber (κ) due to the numerical approximation, i.e. $\bar{f}' = j\bar{\kappa}\bar{f}$ and $\bar{f}' = j\bar{\kappa}\bar{f}$ from Eq. (2.11). The intermediary functions $A_i(\kappa)$, $B_i(\kappa)$, $C_i(\kappa)$ and $D_i(\kappa)$ in Eq. (2.12) are given by

$$i = 0 : \quad \begin{cases} A_0(\kappa) = 1 + \gamma_{01} \cos(\kappa) + \gamma_{02} \cos(2\kappa), \\ B_0(\kappa) = \gamma_{01} \sin(\kappa) + \gamma_{02} \sin(2\kappa), \\ C_0(\kappa) = \sum_{m=-3, \neq 0}^4 b_{0m} [\cos(m\kappa) - 1], \\ D_0(\kappa) = \sum_{m=-3, \neq 0}^4 b_{0m} \sin(m\kappa), \end{cases} \quad (2.13)$$

$$i = 1 : \quad \begin{cases} A_1(\kappa) = 1 + (\gamma_{10} + \gamma_{12}) \cos(\kappa) + \gamma_{13} \cos(2\kappa), \\ B_1(\kappa) = (\gamma_{12} - \gamma_{10}) \sin(\kappa) + \gamma_{13} \sin(2\kappa), \\ C_1(\kappa) = \sum_{m=-3, \neq 1}^4 b_{1m} \{\cos[(m-1)\kappa] - 1\}, \\ D_1(\kappa) = \sum_{m=-3, \neq 1}^4 b_{1m} \sin[(m-1)\kappa]. \end{cases} \quad (2.14)$$

By working out Eq. (2.12), the pseudo-wavenumber of each scheme turns out to be a function of the true wavenumber as follows:

$$\begin{aligned} \bar{\kappa}_i(\kappa) = & \frac{A_i(\kappa)D_i(\kappa) - B_i(\kappa)C_i(\kappa)}{A_i^2(\kappa) + B_i^2(\kappa)} - j \\ & \times \frac{A_i(\kappa)C_i(\kappa) + B_i(\kappa)D_i(\kappa)}{A_i^2(\kappa) + B_i^2(\kappa)} \quad \text{for } i = 0, 1 \end{aligned} \quad (2.15)$$

which is a complex function composed of real ($\text{Re}[\bar{\kappa}_i(\kappa)]$, dispersive) and imaginary ($\text{Im}[\bar{\kappa}_i(\kappa)]$, dissipative) parts. Following [5], the pseudo-wavenumber of the interior central scheme given by Eq. (2.1) is derived as

$$\bar{\kappa}_M(\kappa) = \frac{2 \sum_{m=1}^3 a_m \sin(m\kappa)}{1 + 2\alpha \cos(\kappa) + 2\beta \cos(2\kappa)} \quad \text{for } 2 \leq i \leq N-2 \quad (2.16)$$

which has a real part only contrary to Eq. (2.15).

As indicated earlier, the control variables (ϕ_1, ϕ_2, ϕ_3) in Eqs. (2.2) and (2.3) need to be determined in an optimal sense to finally obtain the coefficients in Eqs. (2.9) and (2.10). The idea of optimization considered in this paper is to minimize the gap between the boundary schemes' $\bar{\kappa}_i(\kappa)$ and the interior scheme's $\bar{\kappa}_M(\kappa)$ within the range of $\kappa \in [0, \pi]$. In other words, the current optimization aims at $\text{Re}[\bar{\kappa}_i(\kappa)] \rightarrow \bar{\kappa}_M(\kappa)$ and $\text{Im}[\bar{\kappa}_i(\kappa)] \rightarrow 0$. In this way, the actual behavior of the boundary schemes may resemble that of the interior scheme, which means that any travelling waves from the interior domain suffer little change in resolution characteristics across the subdomain boundaries. The enhanced uniformity of resolution characteristics helps suppress unwanted dispersive errors that might arise locally.

In order to quantify the discrepancies of pseudo-wavenumber profiles between $\bar{\kappa}_i(\kappa)$ and $\bar{\kappa}_M(\kappa)$, an error function in an L_2 -norm fashion may be defined as follows:

$$\begin{aligned} E(\phi_1, \phi_2, \phi_3) = & \delta \sum_{i=0}^1 \int_0^\pi \{\text{Re}[\bar{\kappa}_i(\kappa)] - \bar{\kappa}_M(\kappa)\}^2 d\kappa + (1 - \delta) \\ & \times \sum_{i=0}^1 \int_0^\pi \text{Im}[\bar{\kappa}_i(\kappa)]^2 d\kappa \end{aligned} \quad (2.17)$$

where δ is a factor to change the weight between the dispersive and the dissipative error. In the current work, $\delta = \pi/(\pi + 1)$ which puts

Table 2
Coefficients for Eqs. (2.9) and (2.10).

Coefficients	$i = 0$	$i = 1$
γ_{i0}	(n/a)	0.3541705155853574
γ_{i1}	0.1515009062303691e+1	(n/a)
γ_{i2}	0.5482503746903026	0.7660970147374699
γ_{i3}	(n/a)	0.1490676611277856
$b_{i(-3)}$: 3rd halo	-0.1063330835808966e-1	-0.8269079858546097e-3
$b_{i(-2)}$: 2nd halo	0.7971337463783268e-1	0.7649269203233134e-2
$b_{i(-1)}$: 1st halo	-0.4054627699009952	-0.6960041907084386e-1
b_{i0}	(n/a)	-0.6742312911913867
b_{i1}	0.3894021071906670	(n/a)
b_{i2}	0.1079836286882313e+1	0.6698644443328796
b_{i3}	0.8280460871175586e-1	0.3848785547254968
b_{i4}	-0.3041253949041069e-2	0.1197389466664873e-1

higher weighting on the dispersive error is chosen after numerous tests for the best outcome. The value of δ has been deduced based on π , since the integral range is $[0, \pi]$ and any profile of $\text{Re}[\bar{\kappa}_i(\kappa)]$ yields a constant error of π at $\kappa = \pi$. An iterative root-finding method such as Newton–Raphson may be used to find a local minimum of the error function:

$$\frac{\partial E}{\partial \phi_1} = 0, \quad \frac{\partial E}{\partial \phi_2} = 0 \quad \text{and} \quad \frac{\partial E}{\partial \phi_3} = 0. \quad (2.18)$$

The solution of Eq. (2.18) depends greatly on the initial guess of $(\phi_1, \phi_2, \phi_3) = (\phi_o, 2\phi_o, 3\phi_o)$ since there exist multiple roots due to the highly nonlinear nature of Eq. (2.17). After experimenting with a variety of the initial guess ϕ_o and checking the roots obtained, an effective set of the control variables (producing the best results in actual computation) is selected as

$$(\phi_1, \phi_2, \phi_3) = (0.095\pi, 0.190\pi, 0.967\pi) \quad (2.19)$$

which provides the final coefficients for Eqs. (2.9) and (2.10) as listed in Table 2. The resulting pseudo-wavenumber profiles from Eq. (2.15) are shown in Fig. 1 in comparison with the target profile of Eq. (2.16). Although the boundary schemes appear to have slightly lower resolution than the interior scheme as shown in Fig. 1, the overall level of errors in actual parallel calculations does not rise to a considerable extent as demonstrated in Sections 6 and 7.

3. Implementation of compact filter for parallel computing

The performance of numerical filters is one of the key elements for a successful solution of DNS/LES/CAA especially when

high-order schemes are concerned. In order to ensure numerical stability and remove non-physical oscillatory components in the solution that usually arise from the irresolvable wavenumber range, the compact finite difference scheme proposed in Section 2 is accompanied by a compact discrete filter in the present work. In particular, a predictor–corrector type implementation of the compact filter is proposed for MPI based parallel computing purposes.

3.1. Variable compact discrete filter

A sixth-order central compact filter with variable coefficients [10] is employed for this work, which also has a pentadiagonal platform and a seven-point stencil similar to Eq. (2.1) as

$$\beta_F \hat{\Delta} f_{i-2} + \alpha_F \hat{\Delta} f_{i-1} + \hat{\Delta} f_i + \alpha_F \hat{\Delta} f_{i+1} + \beta_F \hat{\Delta} f_{i+2} = \sum_{m=1}^3 a_{Fm} (f_{i-m} - 2f_i + f_{i+m}) \quad \text{for } 3 \leq i \leq N-3 \quad (3.1)$$

where “ $\hat{\Delta}$ ” represents numerically filtered quantities and $\hat{\Delta} f_i = \hat{f}_i - f_i$ is the difference between the filtered and the original functions. The filter coefficients are designed to achieve a user-defined cut-off wavenumber (κ_C) required for the system. They may vary in space particularly near the boundaries in order to enhance the stability or quality of solution as shown in [10]. The filter coefficients may be given by trigonometric functions of κ_C as

$$\begin{aligned} \alpha_F &= -\frac{30 \cos(\kappa_C) + 2 \cos(3\kappa_C)}{A_F(\kappa_C)}, \\ \beta_F &= \frac{18 + 9 \cos(\kappa_C) + 6 \cos(2\kappa_C) - \cos(3\kappa_C)}{2A_F(\kappa_C)}, \\ a_{F1} &= \frac{30}{A_F(\kappa_C)} \cos^4\left(\frac{\kappa_C}{2}\right), \quad a_{F2} = -\frac{2a_{F1}}{5} \quad \text{and} \quad a_{F3} = \frac{a_{F1}}{15}, \\ A_F(\kappa_C) &= 30 - 5 \cos(\kappa_C) + 10 \cos(2\kappa_C) - 3 \cos(3\kappa_C). \end{aligned} \quad (3.2)$$

The Fourier transform of the filter via Eq. (2.11) leads to the following transfer function:

$$T(\kappa) \equiv \frac{\hat{f}(k)}{f(k)} = 1 + \frac{2 \sum_{m=1}^3 a_{Fm} [\cos(m\kappa) - 1]}{1 + 2\alpha_F \cos(\kappa) + 2\beta_F \cos(2\kappa)} \quad (3.3)$$

which represents the ratio of the filtered value to the original one at a certain wavenumber, where $T(\kappa) = 1$ is no filtering and $T(\kappa) = 0$ is complete filtering. The cut-off wavenumber is defined to yield $T(\kappa_C) = 1/2$. Various profiles of the transfer function with different cut-off wavenumbers are shown in Fig. 2. The transfer functions are strictly between zero and unity when the cut-off wavenumber

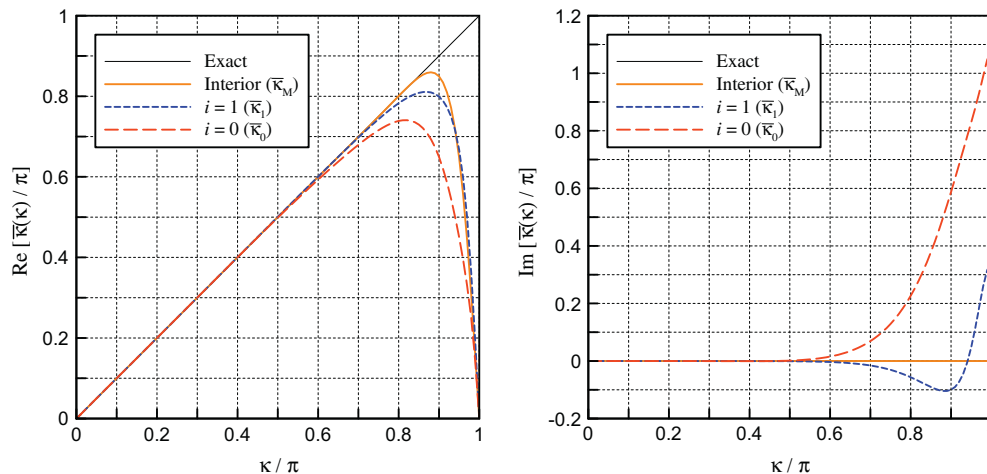


Fig. 1. Pseudo-wavenumber profiles of compact finite difference schemes used in the present work: real (left) and imaginary (right) parts.

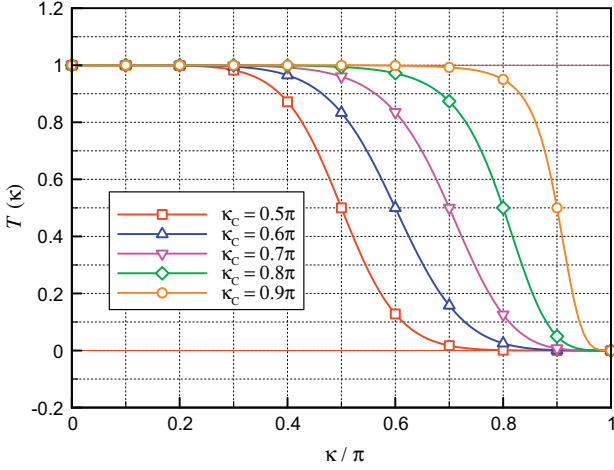


Fig. 2. Profiles of transfer function by Eq. (3.3) for compact filters of Eq. (3.1) with various cut-off wavenumbers.

is within $\pi/2 \leq \kappa_c \leq \pi$. In the current work, $\kappa_c = 0.87\pi$ is selected for Eq. (3.1) as it is found most effective in removing spurious errors as shown in Section 6.3. The resulting coefficients are listed in Table 3. The overall fourth-order accuracy from Eqs. (2.1), (2.9), and (2.10) is unaffected by adopting the sixth-order filter, which is shown in Section 4. Adding a comment, it might be necessary to consider that the most effective cut-off wavenumber may be different in other types of problems not dominated by vortical flows.

3.2. Predictor–corrector steps for parallel implementation

The ability to achieve a high cut-off wavenumber (e.g. $\kappa_c = 0.87\pi$) comes at the cost of having to solve a pentadiagonal matrix system that, analogous to the system of finite difference schemes in Section 2, represents a challenge for efficient computations on massively parallel computers. It was initially considered to develop non-central boundary filters similar to the boundary differencing schemes proposed in the previous section. However, the authors learned instead that implementing the central filter of Eq. (3.1) across subdomain boundaries based on predictor–corrector steps could be remarkably more effective than the old approach without a risk of numerical instability. The new parallel filtering strategy reduces artifact acoustic reflections from subdomain boundaries at a reasonable extra cost. The following steps illustrate the proposed filtering procedure incorporating the predictor–corrector steps:

- A. Initial guess on $\hat{\Delta}f_i^{(0)} = 0$ at halo points ($i = -1$ and -2).
- B. Predictor ($n = 1$) and corrector ($n = 2$) steps (solving for $\hat{\Delta}f_i^{(n)}$).

$$i = 0: \quad \hat{\Delta}f_0^{(n)} + \ddot{\alpha}_F \hat{\Delta}f_1^{(n)} + \ddot{\beta}_{Fm} \hat{\Delta}f_2^{(n)} = \sum_{m=1}^3 \ddot{\alpha}_{Fm} (f_{-m} - 2f_0 + f_m) - \ddot{\alpha}_F \hat{\Delta}f_{-1}^{(n-1)} - \ddot{\beta}_F \hat{\Delta}f_{-2}^{(n-1)} \quad (3.4)$$

$$i = 1: \quad \ddot{\alpha}_F \hat{\Delta}f_0^{(n)} + \hat{\Delta}f_1^{(n)} + \ddot{\alpha}_F \hat{\Delta}f_2^{(n)} + \ddot{\beta}_F \hat{\Delta}f_3^{(n)} = \sum_{m=1}^3 \ddot{a}_{Fm} (f_{1-m} - 2f_1 + f_{1+m}) - \ddot{\beta}_F \hat{\Delta}f_{-1}^{(n-1)} \quad (3.5)$$

Table 3
Coefficients for Eq. (3.1) based on $\kappa_c = 0.87\pi$.

α_F	β_F	a_{F1}	a_{F2}	a_{F3}
0.6645420561839883	0.1670915887632024	0.1195093396506611e-2	-0.4780373586026445e-3	0.7967289310044074e-4

$$i = 2: \quad \dot{\beta}_F \hat{\Delta}f_0^{(n)} + \dot{\alpha}_F \hat{\Delta}f_1^{(n)} + \hat{\Delta}f_2^{(n)} + \dot{\alpha}_F \hat{\Delta}f_3^{(n)} + \dot{\beta}_F \hat{\Delta}f_4^{(n)} = \sum_{m=1}^3 \dot{a}_{Fm} (f_{2-m} - 2f_2 + f_{2+m}) \quad (3.6)$$

$i \geq 3$: implementing Eq. (3.1),

C. Update of filtered function profile ($\hat{f}_i = f_i + \hat{\Delta}f_i^{(2)}$).

This arrangement enables to split a single pentadiagonal matrix system across the entire domain into a number of smaller ones in each subdomain, and to solve them in an iterative manner to approximate the original solution. A higher number of iterations in the predictor step will provide a solution closer to the original one. However, the minimal iteration number ($n = 2$) is suggested to avoid a substantial increase in computational time. In the present work, the extra computational cost due to the iteration (predictor step) turns out to be small because the filter is implemented only at the final stage of the time marching (Runge–Kutta) schemes used.

It should be noted that the filter coefficients in Eqs. (3.4)–(3.6) may differ from those in Eq. (3.1) (as indicated by “.”, “..” and “...” on the coefficients) via readjusting the cut-off wavenumber in Eq. (3.2). Despite the optimization carried out in Section 2.3, the resolution characteristics of the boundary differencing schemes still show some deviation from those of the interior scheme as seen in Fig. 1. Due to the non-uniform resolution characteristics, it may be necessary to readjust the filter cut-off wavenumber locally around the boundaries. The authors suggest a smooth profile of the cut-off wavenumber around the boundaries as

$$\begin{aligned} \ddot{\kappa}_c &= \kappa_c + (\kappa_c^* - \kappa_c) \sin^2(3\pi/7) \quad \text{for } i = 0, \\ \dot{\kappa}_c &= \kappa_c + (\kappa_c^* - \kappa_c) \sin^2(2\pi/7) \quad \text{for } i = 1, \\ \dot{\kappa}_c &= \kappa_c + (\kappa_c^* - \kappa_c) \sin^2(\pi/7) \quad \text{for } i = 2, \end{aligned} \quad (3.7)$$

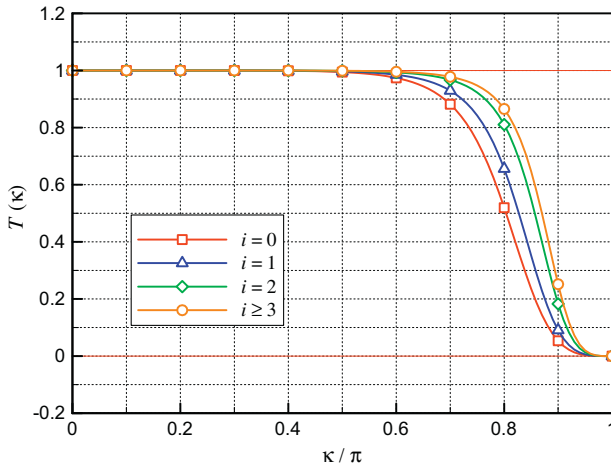
which varies from the assigned value of $\kappa_c = 0.87\pi$ in the interior domain to a certain value close to κ_c^* at the boundary. It is ensured that the profile of Eq. (3.7) is symmetric without discontinuity across the boundary to keep the solution smooth, i.e. $\sin[(3-i)\pi/7]$ is symmetric from $i = -1/2$ where $i = -1/2$ represents the middle of the boundary and the first halo point. This arrangement is aimed to compensate the non-uniform behaviors of Eqs. (2.9) and (2.10) and reduce errors arising from the subdomain boundaries. It is shown in Section 6.3 that $\kappa_c^* = 0.8\pi$ gives best results for the present differencing–filtering system. These cut-off wavenumbers may change if a different compact scheme and/or coefficients are chosen instead of those suggested in Section 2. The resulting filter coefficients based on Eq. (3.2) and (3.7) are listed in Table 4, and the transfer function profiles are plotted in Fig. 3.

4. Grid convergence test and stability analysis

This section examines the grid convergence rate and numerical stability of the proposed compact differencing–filtering system in a parallel computing setup. The grid convergence test aims to check the formal orders of accuracy in actual computation, which are designed to be fourth-order in differencing and sixth-order in filtering. The numerical stability is assessed by a classical eigenvalue analysis approach based on a one-dimensional linear scalar wave equation.

Table 4Coefficients for Eqs. (3.4)–(3.6) based on $\kappa_C = 0.87\pi$ and $\kappa_C^* = 0.8\pi$.

$i = 0$	$i = 1$	$i = 2$
$\kappa_C = 0.8034660896234154\pi$	$\kappa_C = 0.8272117673115290\pi$	$\kappa_C = 0.8568221430650557\pi$
$\alpha_F = 0.6542278901742072$	$\alpha_F = 0.6595604150977900$	$\alpha_F = 0.6634786817222235$
$\beta_F = 0.1691544219651586$	$\beta_F = 0.1680879169804420$	$\beta_F = 0.1673042636555553$
$\bar{a}_{F1} = 0.6996811777008479e - 2$	$\bar{a}_{F1} = 0.3997266507493101e - 2$	$\bar{a}_{F1} = 0.1793241531249329e - 2$
$\bar{a}_{F2} = -0.2798724710803392e - 2$	$\bar{a}_{F2} = -0.1598906602997240e - 2$	$\bar{a}_{F2} = -0.7172966124997318e - 3$
$\bar{a}_{F3} = 0.4664541184672319e - 3$	$\bar{a}_{F3} = 0.2664844338328734e - 3$	$\bar{a}_{F3} = 0.1195494354166220e - 3$

**Fig. 3.** Profiles of transfer function for the current compact filters based on the coefficients given in Tables 3 and 4.

4.1. Grid convergence test

The grid convergence test of the proposed compact finite difference scheme and filter is based on the following test function:

$$f(x) = f_\infty [2 + \cos(k_1 x/L)] \exp[-(\ln 2)(k_2 x/L)^2] \quad \text{for } x \in [L/2, L/2] \quad (4.1)$$

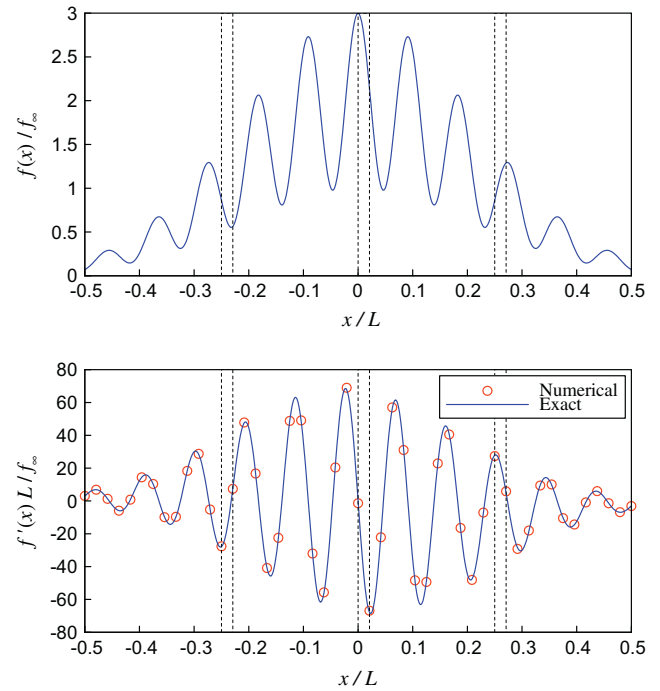
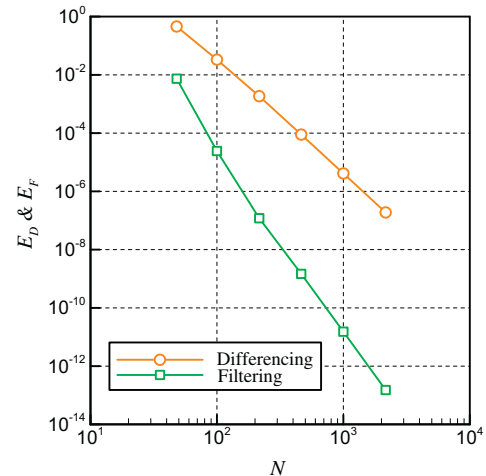
where the constants are set to $k_1 = 17k_2$ and $k_2 = 4$. The domain is split into four subdomains: $x \in [-L/2, -L/4]$, $[-L/4 + \Delta x, 0]$, $[\Delta x, L/4]$ and $[L/4 + \Delta x, L/2]$, which have $N/4 + 1$, $N/4$, $N/4$ and $N/4$ number of grid points, respectively ($N + 1$ in total), where $i = 0$ and $i = N$ represent the left and the right end of the domain, respectively. The grid points are equally spaced, i.e. $\Delta x = L/N$. The proposed compact finite difference scheme consisting of Eqs. (2.1), (2.9), and (2.10), and the filter of Eqs. (3.1) and (3.4)–(3.6) are implemented in each subdomain. The exact values of Eq. (4.1) are specified for the virtual boundary conditions beyond the domain ends ($x < -L/2$ and $x > L/2$), where the contribution of the filter is set to zero ($\hat{\Delta}f_i = 0$ for $i = -2, -1, N + 1$ and $N + 2$) in addition.

Fig. 4 shows the profiles of Eq. (4.1) and its first derivative in comparison with numerical data obtained by using the proposed compact scheme in a parallel manner across the four subdomains. The numerical errors due to the differencing and filtering are defined on L_∞ -norm basis in a dimensionless form as

$$E_D = \max_i |\bar{f}'_i - f'(x_i)| / (L/f_{\max}), \quad (4.2)$$

$$E_F = \max_i |\Delta \hat{f}_i| / f_{\max},$$

which is intended to focus on peak errors from the subdomain boundaries. This is a more stringent measure of errors compared to conventional L_2 -norm based ones. Fig. 5 confirms the fourth-order accuracy of differencing and the sixth-order of filtering as

**Fig. 4.** Test function given by Eq. (4.1) (top) and its first derivative (bottom) calculated by the proposed compact finite difference scheme ($N = 48$). Vertical dashed lines represent the location of subdomain boundaries.**Fig. 5.** Grid convergence test on the proposed compact finite difference scheme (E_D) and filter (E_F) based on Eq. (4.2).

expected despite the strict definition of errors. This shows that the proposed parallel computing approach maintains the original level of accuracy.

4.2. Linear stability analysis

The stability analysis of the proposed differencing-filtering system begins by considering a one-dimensional linear scalar wave equation:

$$\frac{\partial f}{\partial t} + c_\infty \frac{\partial f}{\partial x} = 0 \quad (4.3)$$

over a domain $x \in [0, L]$ where c_∞ is the wave convection speed ($c_\infty > 0$). Complying with the direction of the wave, a prescribed boundary condition is required in the upstream domain ($x < 0$). Current setup for the stability analysis is that the proposed boundary schemes and filters are used at the left boundary ($x = 0$), and the upstream boundary condition is provided by halo cells located outside the domain. Conventional boundary schemes [5] and filters [10] are used at the right boundary ($x = L$) through which the prescribed wave leaves the domain. For the purpose of stability analysis, the upstream boundary condition can be set to zero, $f_{BC}(x < 0, t) = 0$ without loss of generality [12]. The domain is discretized into N equal intervals with $\Delta x = L/N$, which leaves $N + 1$ unknowns to be found ($i = 0, \dots, N$).

The spatial derivative $\partial f / \partial x$ in Eq. (4.3) numerically evaluated by the compact finite difference scheme yields a linear system of equations expressed in a matrix–vector form as

$$\mathbf{P}\mathbf{f}' = \frac{1}{\Delta x} \mathbf{Q}(\mathbf{f} + \sigma \hat{\Delta} \mathbf{f}) \quad \text{for } \sigma = \begin{cases} 0: & \text{without filter} \\ 1: & \text{with filter} \end{cases} \quad (4.4)$$

where $\mathbf{f}$, $\hat{\Delta} \mathbf{f}$ and $\mathbf{f}'$ are N -dimensional vectors representing the nodal values of the objective function, the filter's contribution and the numerical derivative, respectively:

$$\mathbf{f} = (f_0, \dots, f_N)^T, \quad \hat{\Delta} \mathbf{f} = (\hat{\Delta} f_0, \dots, \hat{\Delta} f_N)^T \quad \text{and} \quad \mathbf{f}' = (\bar{f}'_0, \dots, \bar{f}'_N)^T.$$

The $N \times N$ matrices $\mathbf{P}$ and $\mathbf{Q}$ in Eq. (4.4) are as follows:

$$\mathbf{P} = \begin{pmatrix} 1 & \gamma_{01} & \gamma_{02} & 0 & \dots & 0 & 0 & 0 & 0 \\ \gamma_{10} & 1 & \gamma_{12} & \gamma_{13} & 0 & \dots & 0 & 0 & 0 \\ \beta & \alpha & 1 & \alpha & \beta & 0 & \dots & 0 & 0 \\ 0 & \beta & \alpha & 1 & \alpha & \beta & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & \beta & \alpha & 1 & \alpha & \beta & 0 \\ 0 & 0 & \dots & 0 & \hat{\gamma}_{24} & \hat{\gamma}_{23} & 1 & \hat{\gamma}_{21} & \hat{\gamma}_{20} \\ 0 & 0 & 0 & \dots & 0 & \hat{\gamma}_{13} & \hat{\gamma}_{12} & 1 & \hat{\gamma}_{10} \\ 0 & 0 & 0 & 0 & \dots & 0 & \hat{\gamma}_{02} & \hat{\gamma}_{01} & 1 \end{pmatrix}, \quad (4.5)$$

$$\mathbf{Q} = \begin{pmatrix} b_{00} & b_{01} & b_{02} & b_{03} & b_{04} & 0 & 0 & 0 & 0 & \dots & 0 \\ b_{10} & b_{11} & b_{12} & b_{13} & b_{14} & 0 & 0 & 0 & 0 & \dots & 0 \\ -a_2 & -a_1 & 0 & a_1 & a_2 & a_3 & 0 & 0 & 0 & \dots & 0 \\ -a_3 & -a_2 & -a_1 & 0 & a_1 & a_2 & a_3 & 0 & 0 & \dots & 0 \\ 0 & -a_3 & -a_2 & -a_1 & 0 & a_1 & a_2 & a_3 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & -a_3 & -a_2 & -a_1 & 0 & a_1 & a_2 & a_3 & 0 \\ 0 & \dots & 0 & 0 & -a_3 & -a_2 & -a_1 & 0 & a_1 & a_2 & a_3 \\ 0 & \dots & 0 & 0 & -\hat{b}_{26} & -\hat{b}_{25} & -\hat{b}_{24} & -\hat{b}_{23} & -\hat{b}_{22} & -\hat{b}_{21} & -\hat{b}_{20} \\ 0 & \dots & 0 & 0 & -\hat{b}_{16} & -\hat{b}_{15} & -\hat{b}_{14} & -\hat{b}_{13} & -\hat{b}_{12} & -\hat{b}_{11} & -\hat{b}_{10} \\ 0 & \dots & 0 & 0 & -\hat{b}_{06} & -\hat{b}_{05} & -\hat{b}_{04} & -\hat{b}_{03} & -\hat{b}_{02} & -\hat{b}_{01} & -\hat{b}_{00} \end{pmatrix} \quad (4.6)$$

where the coefficients are given by Tables 1 and 2, and those with a round hat “ $\hat{\cdot}$ ” used for general boundaries can be found in [5]. The coefficients at the diagonal of Eq. (4.6) are defined as

$$b_{00} = - \sum_{m=-3, \neq 0}^4 b_{0m} \quad \text{and} \quad b_{11} = - \sum_{m=-3, \neq 1}^4 b_{1m}. \quad (4.7)$$

Similarly, the filter's contribution in Eq. (4.4) can be calculated by solving the following equation comprising Eqs. (3.1) and (3.4)–(3.6):

$$\mathbf{R}\hat{\Delta} \mathbf{f} = \mathbf{S}\mathbf{f} \quad (4.8)$$

with the $N \times N$ matrices $\mathbf{R}$ and $\mathbf{S}$ expressed by

$$\mathbf{R} = \begin{pmatrix} 1 & \ddot{\alpha}_F & \ddot{\beta}_F & 0 & \dots & 0 & 0 & 0 & 0 \\ \ddot{\alpha}_F & 1 & \ddot{\alpha}_F & \ddot{\beta}_F & 0 & \dots & 0 & 0 & 0 \\ \ddot{\beta}_F & \ddot{\alpha}_F & 1 & \ddot{\alpha}_F & \ddot{\beta}_F & 0 & \dots & 0 & 0 \\ 0 & \beta_F & \alpha_F & 1 & \alpha_F & \beta_F & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & \beta_F & \alpha_F & 1 & \alpha_F & \beta_F & 0 \\ 0 & 0 & \dots & 0 & \hat{\gamma}_{F24} & \hat{\gamma}_{F23} & 1 & \hat{\gamma}_{F21} & \hat{\gamma}_{F20} \\ 0 & 0 & 0 & \dots & 0 & \hat{\gamma}_{F13} & \hat{\gamma}_{F12} & 1 & \hat{\gamma}_{F10} \\ 0 & 0 & 0 & 0 & \dots & 0 & \hat{\gamma}_{F02} & \hat{\gamma}_{F01} & 1 \end{pmatrix}, \quad (4.9)$$

$$\mathbf{S} = \begin{pmatrix} \ddot{a}_{F0} & \ddot{a}_{F1} & \ddot{a}_{F2} & \ddot{a}_{F3} & 0 & 0 & 0 & 0 & 0 & \dots & 0 \\ \ddot{a}_{F1} & \ddot{a}_{F0} & \ddot{a}_{F1} & \ddot{a}_{F2} & \ddot{a}_{F3} & 0 & 0 & 0 & 0 & \dots & 0 \\ \ddot{a}_{F2} & \ddot{a}_{F2} & \ddot{a}_{F0} & \ddot{a}_{F1} & \ddot{a}_{F2} & \ddot{a}_{F3} & 0 & 0 & 0 & \dots & 0 \\ \ddot{a}_{F3} & \ddot{a}_{F2} & \ddot{a}_{F1} & \ddot{a}_{F0} & \ddot{a}_{F1} & \ddot{a}_{F2} & \ddot{a}_{F3} & 0 & 0 & \dots & 0 \\ 0 & \ddot{a}_{F3} & \ddot{a}_{F2} & \ddot{a}_{F1} & \ddot{a}_{F0} & \ddot{a}_{F1} & \ddot{a}_{F2} & \ddot{a}_{F3} & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & \ddot{a}_{F3} & \ddot{a}_{F2} & \ddot{a}_{F1} & \ddot{a}_{F0} & \ddot{a}_{F1} & \ddot{a}_{F2} & \ddot{a}_{F3} & 0 \\ 0 & \dots & 0 & 0 & \ddot{a}_{F3} & \ddot{a}_{F2} & \ddot{a}_{F1} & \ddot{a}_{F0} & \ddot{a}_{F1} & \ddot{a}_{F2} & \ddot{a}_{F3} \\ 0 & \dots & 0 & 0 & 0 & \hat{b}_{F25} & \hat{b}_{F24} & \hat{b}_{F23} & \hat{b}_{F22} & \hat{b}_{F21} & \hat{b}_{F20} \\ 0 & \dots & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \dots & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (4.10)$$

where the coefficients are given by Tables 3 and 4, and the coefficients with round hat “ $\hat{\cdot}$ ” can be found in [10], which were obtained for general boundaries. The coefficients at the diagonal of Eq. (4.10) are defined as

$$(a_{F0}, \hat{a}_{F0}, \ddot{a}_{F0}, \ddot{\ddot{a}}_{F0}) = -2 \sum_{m=1}^3 (a_{Fm}, \hat{a}_{Fm}, \ddot{a}_{Fm}, \ddot{\ddot{a}}_{Fm}). \quad (4.11)$$

Substituting Eq. (4.8) into Eq. (4.4) leads to

$$\mathbf{P}\bar{\mathbf{f}}' = \frac{1}{\Delta x} \mathbf{Q}(\mathbf{I} + \sigma \mathbf{R}^{-1} \mathbf{S})\mathbf{f} \quad \text{for } \sigma = \begin{cases} 0: & \text{without filter} \\ 1: & \text{with filter} \end{cases} \quad (4.12)$$

where $\mathbf{I}$ is an identity matrix. Applying the numerical differentiation and filtering described as above to Eq. (4.3) then leads to a matrix–vector form:

$$\mathbf{P} \frac{d\mathbf{f}}{dt} = -\frac{c_\infty}{\Delta x} \mathbf{Q}(\mathbf{I} + \sigma \mathbf{R}^{-1} \mathbf{S})\mathbf{f}. \quad (4.13)$$

Since Eq. (4.13) is a system of ordinary differential equations in time with constant coefficients the solution consists of normal modes $\mathbf{f} = \mathbf{f}_0 e^{w t}$ with a constant w representing the rates of decay or amplification of the modes. Imposing the normal modes into Eq. (4.13) leads to an eigenvalue problem:

$$-\mathbf{P}^{-1} \mathbf{Q}(\mathbf{I} + \sigma \mathbf{R}^{-1} \mathbf{S})\mathbf{f}_0 = \omega \mathbf{f}_0 \quad (4.14)$$

where $\omega = w \Delta x / c_\infty$ is the dimensionless eigenvalue and $\mathbf{f}_0$ becomes the corresponding eigenvector. The real parts of the eigenvalues are

required to be equal to or less than zero to ensure numerical stability of the system, i.e. $|e^{w\Delta t}| \leq 1$. Fig. 6 shows the distribution of eigenvalues in the complex plane with the filtering switched off ($\sigma = 0$) and on ($\sigma = 1$) for four different numbers of nodes specified ($N = 8, 16, 32$ and 64). All the eigenvalues fall on the left half plane, which demonstrates the linear stability of the proposed differencing-filtering system. Moreover, it can be observed that the inclusion of filtering only has a moderate effect on the overall stability of the scheme.

4.3. Calculation of linear wave convection

The one-dimensional linear wave convection in Eq. (4.3) is numerically calculated by using the proposed differencing-filtering system. The initial wave profile is given by Eq. (4.1) with $k_1 = 17k_2$ where various values of k_2 are used for a parametric test. The upstream boundary condition is set to zero for the profile and its derivative as well. The calculation runs within a domain of $x \in [-L, 2L]$ where the center of the wave pulse travels from $x/L = 0$ to 1 over a time period of $c_\infty t/L = 1$. The domain is discretized by $N = 480$ equidistant grid cells ($\Delta x/L = 6.25e-3$) and is split into four subdomains. Time marching of the solution is performed by using a standard fourth-order Runge–Kutta method with the CFL number of 0.5 ($c_\infty \Delta t/L = 3.125e-3$) based on which the calculation completes after 320 time steps.

Fig. 7 shows the results of the calculation (for $k_2 = 10$), which also supports the evidence of numerical stability of the proposed differencing-filtering system in addition to the prior eigenvalue

analysis. The present results are compared with those obtained by using fourth- and sixth-order standard finite difference schemes that yield no additional errors due to parallel implementation. All the cases commonly employed the present compact filters in order to focus on the effects of the finite difference schemes used. It is highlighted in Fig. 7 that the proposed parallel differencing-filtering system performs significantly better than the standard explicit schemes. The consistent superior performance of the present approach is demonstrated in Fig. 8, which shows that the level of errors stays one or two orders of magnitude lower than those of the standard explicit schemes over a range of wavelengths (k_2 is inversely proportional to the wavelength). This outcome implies that artifact errors arising from subdomain boundaries in the proposed parallel computing approach is sufficiently negligible in linear wave propagation problems. However, the errors may become more significant in vortex convection problems as discussed in the later sections.

5. Governing equations

The current calculations are based on the generalized compressible Euler/Navier–Stokes equations in a full conservative form as

$$\frac{\partial \tilde{\mathbf{Q}}}{\partial t} + \frac{\partial (\tilde{\mathbf{E}}_m - \tilde{\mathbf{E}}_{vm})}{\partial \xi_m} = \mathbf{0} \quad \text{for } m = 1, 2, 3 \text{ (in three-dimensions)} \quad (5.1)$$

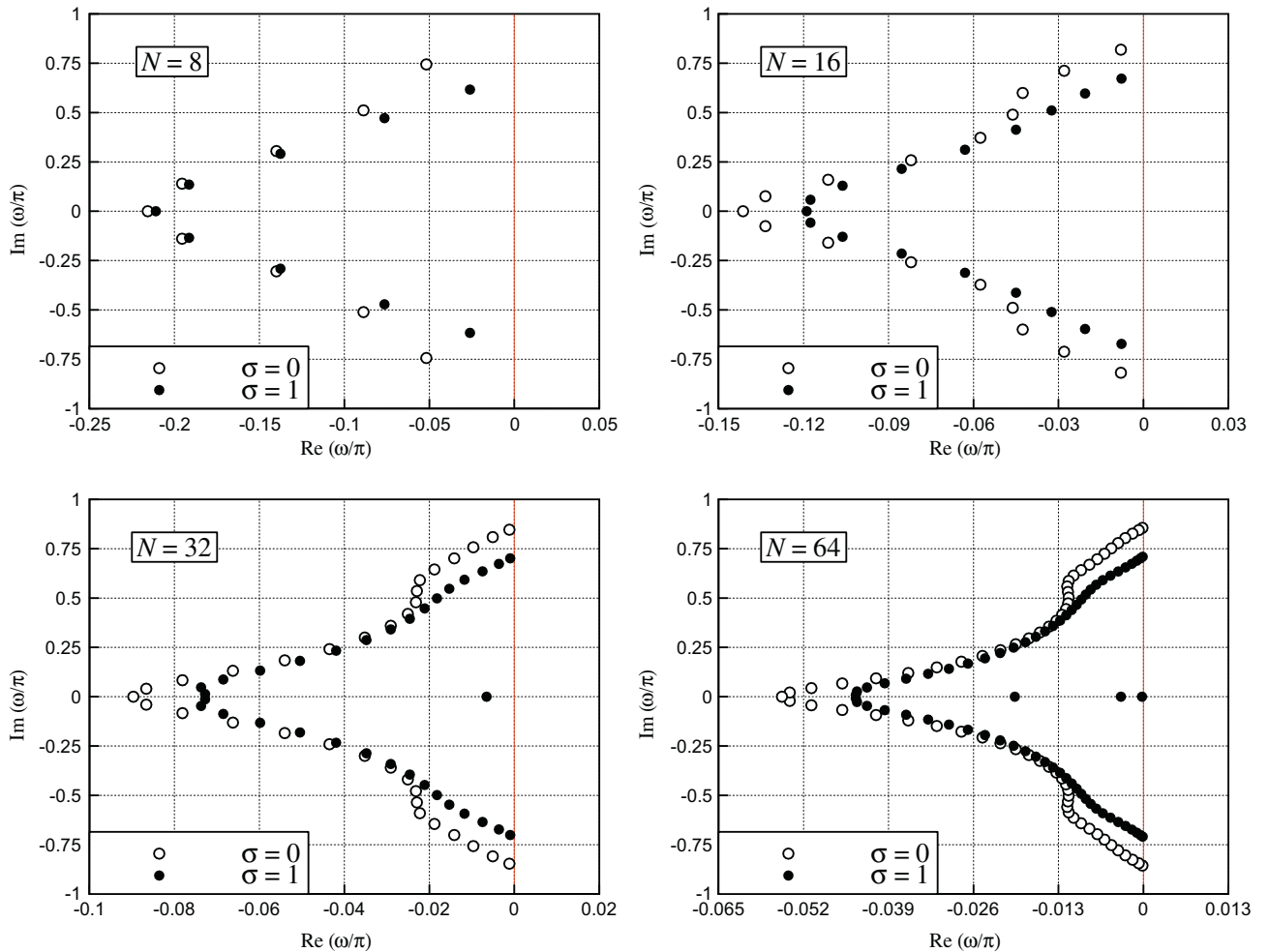


Fig. 6. Distribution of eigenvalues from Eq. (4.14) in four different numbers of grid cells (N) with filtering excluded ($\sigma = 0$) and included ($\sigma = 1$).

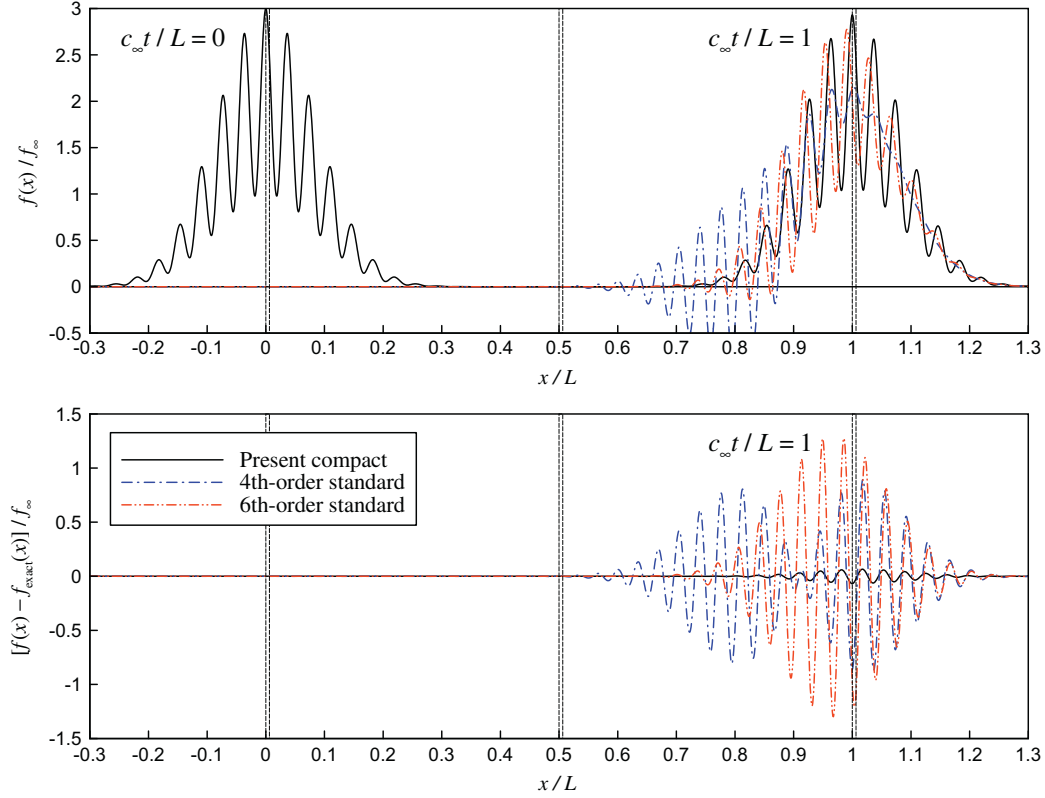


Fig. 7. Calculation of one-dimensional linear wave convection given by Eqs. (4.1) and (4.3) with $k_1 = 17k_2$ and $k_2 = 10$. Solution (top) and error (bottom) profiles. Vertical dashed lines represent the location of subdomain boundaries.

where the tilde represents properties in generalized coordinates, which should not be confused with the Fourier transformation used in Sections 3 and 4. The vectors of conservative variables and fluxes in Eq. (5.1) are given by

$$\tilde{\mathbf{Q}} = \frac{\mathbf{Q}}{J} = \frac{1}{J} \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ \rho e_t \end{pmatrix}, \quad \tilde{\mathbf{E}}_m = \frac{1}{J} \begin{pmatrix} \rho U_m \\ \rho u U_m + \xi_{mx} p \\ \rho v U_m + \xi_{my} p \\ \rho w U_m + \xi_{mz} p \\ (\rho e_t + p) U_m \end{pmatrix} \quad \text{and} \quad (5.2)$$

$$\tilde{\mathbf{E}}_{Vm} = \frac{1}{J} \begin{pmatrix} 0 \\ \xi_{mx} \tau_{xx} + \xi_{my} \tau_{xy} + \xi_{mz} \tau_{xz} \\ \xi_{mx} \tau_{yx} + \xi_{my} \tau_{yy} + \xi_{mz} \tau_{yz} \\ \xi_{mx} \tau_{zx} + \xi_{my} \tau_{zy} + \xi_{mz} \tau_{zz} \\ u \tilde{E}_{V2} + v \tilde{E}_{V3} + w \tilde{E}_{V3} + Q_m \end{pmatrix}$$

where the contravariant velocities, heat fluxes and total energy per unit mass are defined as $U_m = \xi_{mx} u + \xi_{my} v + \xi_{mz} w$, $Q_m = \xi_{mx} q_x + \xi_{my} q_y + \xi_{mz} q_z$ and $e_t = p/[(\gamma - 1)\rho] + (u^2 + v^2 + w^2)/2$. The ratio of specific heats $\gamma = c_p/c_v = 1.4$ is used for air in this paper. The stress tensor components and heat fluxes in Cartesian coordinates are

$$(\tau_{xx}, \tau_{yy}, \tau_{zz}) = \frac{2\mu}{Re_\infty} \left[\left(\frac{\partial u}{\partial x}, \frac{\partial v}{\partial y}, \frac{\partial w}{\partial z} \right) - \frac{1}{3} \nabla \cdot \vec{V} \right],$$

$$\tau_{xy} = \tau_{yx} = \frac{\mu}{Re_\infty} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), \quad \tau_{yz} = \tau_{zy} = \frac{\mu}{Re_\infty} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right),$$

$$\tau_{zx} = \tau_{xz} = \frac{\mu}{Re_\infty} \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right),$$

$$(q_x, q_y, q_z) = \frac{\mu}{(\gamma - 1) Pr Re_\infty M_\infty^2} \left(\frac{\partial T}{\partial x}, \frac{\partial T}{\partial y}, \frac{\partial T}{\partial z} \right).$$

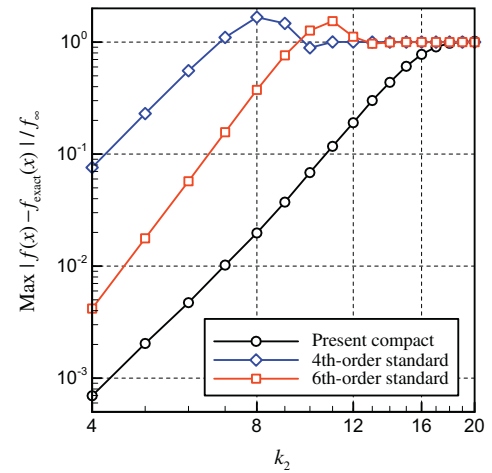


Fig. 8. Maximum error levels in the calculation of one-dimensional linear wave convection varying with k_2 (the wavelength is proportional to $1/k_2$). The plots are based on the results at $c_\infty t/L = 1$.

The dynamic viscosity μ is estimated by using Sutherland's law and a Prandtl number of 0.72 is used in this paper. The grid transformation metrics and the Jacobian follow the conventional definitions.

Time advancing of the solution of Eq. (5.1) is carried out by fourth-order Runge–Kutta methods with the time step size determined by the Courant–Friedrichs–Lewy (CFL) number specified for each case. The CFL number used in this paper is defined as

$$CFL = \max \left[\sum_{m=1}^3 \frac{\Delta t}{\Delta \xi_m} (|U_m| + a \sqrt{\xi_{mx}^2 + \xi_{my}^2 + \xi_{mz}^2}) \right] \quad (5.6)$$

The compact filter is applied to the flow variables ($\mathbf{q}$) only at the last stage of the Runge–Kutta routine in order to minimize the computational cost without degrading the solution as indicated in [13]:

$$\hat{\mathbf{q}} = \mathbf{q} + \hat{\Delta}_{\xi_1} \mathbf{q} + \hat{\Delta}_{\xi_2} \mathbf{q} + \hat{\Delta}_{\xi_3} \mathbf{q} \quad \text{with } \mathbf{q} = (\rho, u, v, w, p)^T \quad (5.5)$$

where $\hat{\Delta}_{\xi_m}$ represents the filtering with respect to the coordinate ξ_m ($m = 1, 2, 3$).

6. Application to Euler calculations

This section demonstrates the performance of the proposed compact differencing-filtering system in parallel computing environments through Euler calculations of two-dimensional isentropic vortex convection which is a standard benchmark problem. The accuracy and efficiency of the current approach are quantified by comparing with the analytic solution and data from single-domain serial calculation.

6.1. Problem definition

The first problem simulates the convection of two-dimensional isentropic vorticity waves in a subsonic free stream. An initial isentropic vorticity wave packet located in the upstream part of the computational domain moves downstream and passes through subdomain boundaries in a parallel computing setup. Additional errors (spurious acoustic reflections) may emerge while the vortex crosses the subdomain boundaries, compared to the serial (single domain) case. The additional errors are carefully investigated by a series of tests and analyses. The isentropic vortex is described by the following equations over a rectangular domain of $x \in [-0.5L, 0.5L]$ and $y \in [-0.5L, 0.5L]$:

$$\left. \begin{aligned} \rho(x, y, t) &= \rho_\infty [1 - \frac{\gamma-1}{2} \psi^2(x^*, y)]^{1/(\gamma-1)} \\ u(x, y, t) &= u_\infty + a_\infty (k_1 y/L) \psi(x^*, y) \\ v(x, y, t) &= -a_\infty (k_1 x^*/L) \psi(x^*, y) \\ p(x, y, t) &= p_\infty (\rho/\rho_\infty)^\gamma \end{aligned} \right\} \quad \text{with } \begin{cases} \psi(x, y) = k_2 \exp[-\frac{1}{2} k_1^2 (x^2 + y^2)/L^2] \\ x^* = x - x_0 - u_\infty t \end{cases} \quad (6.1)$$

The size and the strength of the vortex are controlled by k_1 and k_2 respectively. Various sizes and strengths are tested. Initial location of the vortex core is $(x, y) = (x_0, 0)$ where $x_0 = -0.125L$ is selected in this work. The free stream velocity $u_\infty = M_\infty a_\infty$ is given by the Mach number $M_\infty = 0.5$, where $a_\infty = (\gamma p_\infty / \rho_\infty)^{1/2}$ is the ambient speed of sound. The calculations are carried out on three different (one uniform and two curvilinear) grids with a fixed number of cells $(N \times N) = (200 \times 200)$. The grids are generated by the following formula used by Gaitonde and Visbal [14]:

$$\left. \begin{aligned} x_{ij}/L &= -\frac{1}{2} + \frac{1}{N} [i + \varepsilon \sin(4\pi j/N)] \\ y_{ij}/L &= -\frac{1}{2} + \frac{1}{N} [j + \varepsilon \sin(4\pi i/N)] \end{aligned} \right\} \quad \text{for } 0 \leq i \leq N \quad \text{and} \quad 0 \leq j \leq N \quad (6.2)$$

which gives curvilinear grids deformed from a uniform grid by controlling the constant ε . Setting $\varepsilon = 0$ restores the uniform grid. Four different values of the constant $\varepsilon = 0, 1, 2$ and 3 are tested. Non-reflecting boundary conditions [15] are implemented at the outer boundaries with the time derivatives of all the incoming waves set to zero.

6.2. Results and analysis of data

The domain is split into two (left and right) and then four (top and bottom in addition) subdomains to see the effect of decomposition in

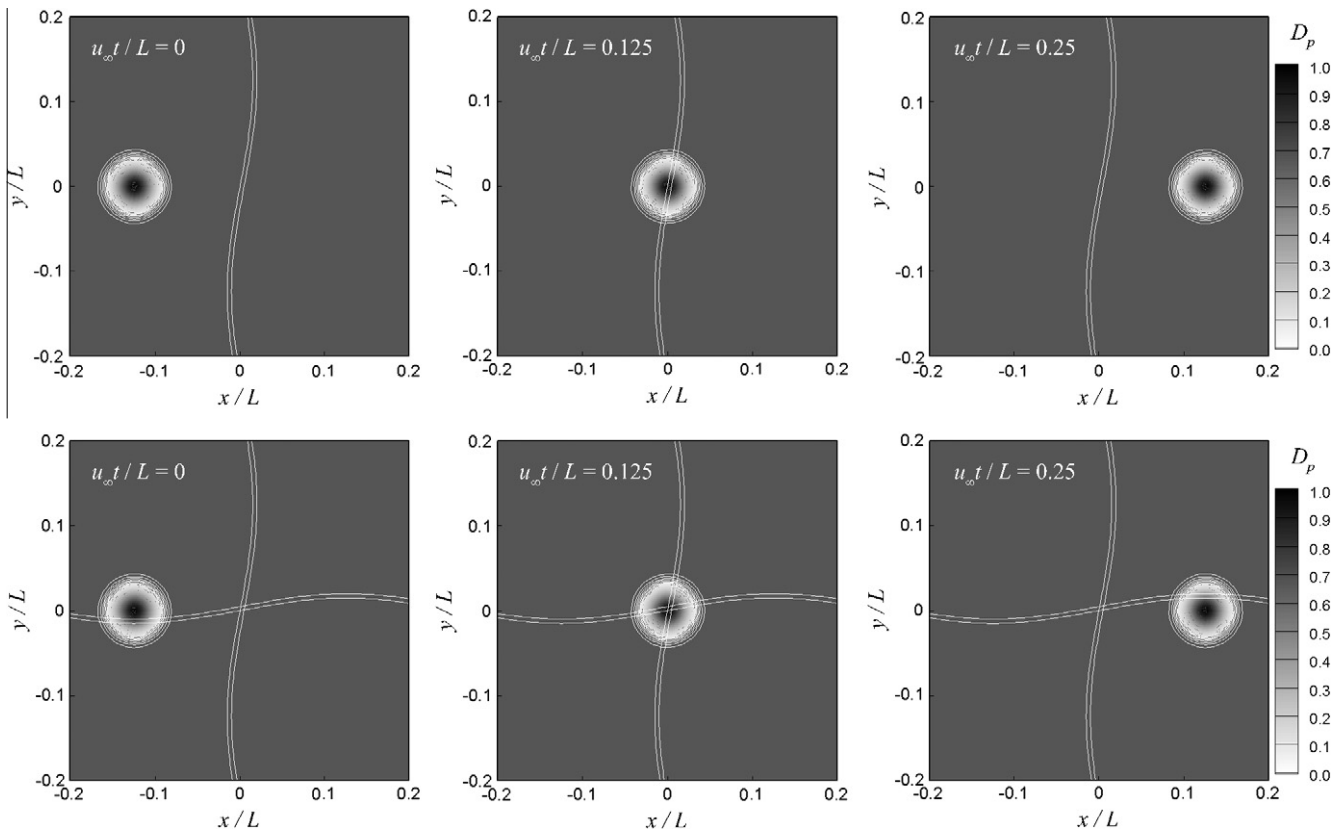


Fig. 9. Contour plots of normalized pressure (D_p) given by Eq. (6.3) in 100 levels from 0.01 to 1. Calculations with $k_1 = 50$ and $k_2 = 0.24$ in Eq. (6.1). Zoomed view: $x \in [-0.2L, 0.2L]$ and $y \in [-0.2L, 0.2L]$. Original domain split into two subdomains (top) and four subdomains (bottom). Curves in the pictures: subdomain boundaries in a curvilinear grid with $\varepsilon = 3$ in Eq. (6.2).

both directions. The calculation continues until $u_{\infty}t/L = 0.25$ by which the vortex has travelled a quarter of the domain length after passing through the subdomain boundaries in the middle. The time step sizes are determined by the CFL number of 0.95. All the results are compared with reference data from single-domain serial calculation and the analytic solution given by Eq. (6.1). In order to examine the accuracy of calculations properly, the following quantity is introduced:

$$D_p = \frac{p_{\infty} - p}{p_{\infty} - p_{\min}} = \frac{1 - p/p_{\infty}}{1 - (1 - \frac{\gamma-1}{2} k_2^2)^{\gamma/(\gamma-1)}} \quad (6.3)$$

which gives the amount of pressure drop from the ambient value normalized by the largest drop that occurs at the vortex core where the pressure is minimum. This normalization leads to the value of D_p to be unity in the vortex core and approaching zero in the free-stream. It is then convenient to define the perimeter of a vortex to be at $D_p = 0.01$ based on which the diameter of the vortex can be precisely measured.

Some snapshots of the D_p profile for a small pseudo-nonlinear vortex ($p_{\min} = 0.9603p_{\infty}$ with 17 cells across the vortex diameter) are plotted in Fig. 9. No noticeable deformation or degradation of

the vortex is found across the subdomain boundaries on a significantly curved grid. In order to examine the results in more detail, the numerical error in pressure is quantified by

$$E_p = |D_p - D_p^{(\text{exact})}| = \frac{|p - p_{\text{exact}}|}{p_{\infty} - p_{\min}}, \quad (6.4)$$

which effectively shows the deviation of pressure from the exact solution normalized by the pressure drop at the vortex core. The distribution of E_p taken at the last moment $u_{\infty}t/L = 0.25$ is plotted in Fig. 10 comparing the two parallel computation cases against the original single-domain serial case. The plots are produced in logarithmic scales to distinguish the order of magnitudes of the errors. The left and middle columns of Fig. 10 show that the dominant errors are located in the vicinity of the vortex core and they are at least three orders of magnitude smaller than the core pressure drop in both the serial and the parallel cases. The right column of Fig. 10 reveals additional errors (acoustic reflections) from the subdomain boundaries. The acoustic reflection (up to $4.0 \times 10^{-6}p_{\infty}$ in Fig. 10) turns out to be one order of magnitude smaller than the existing peak errors.

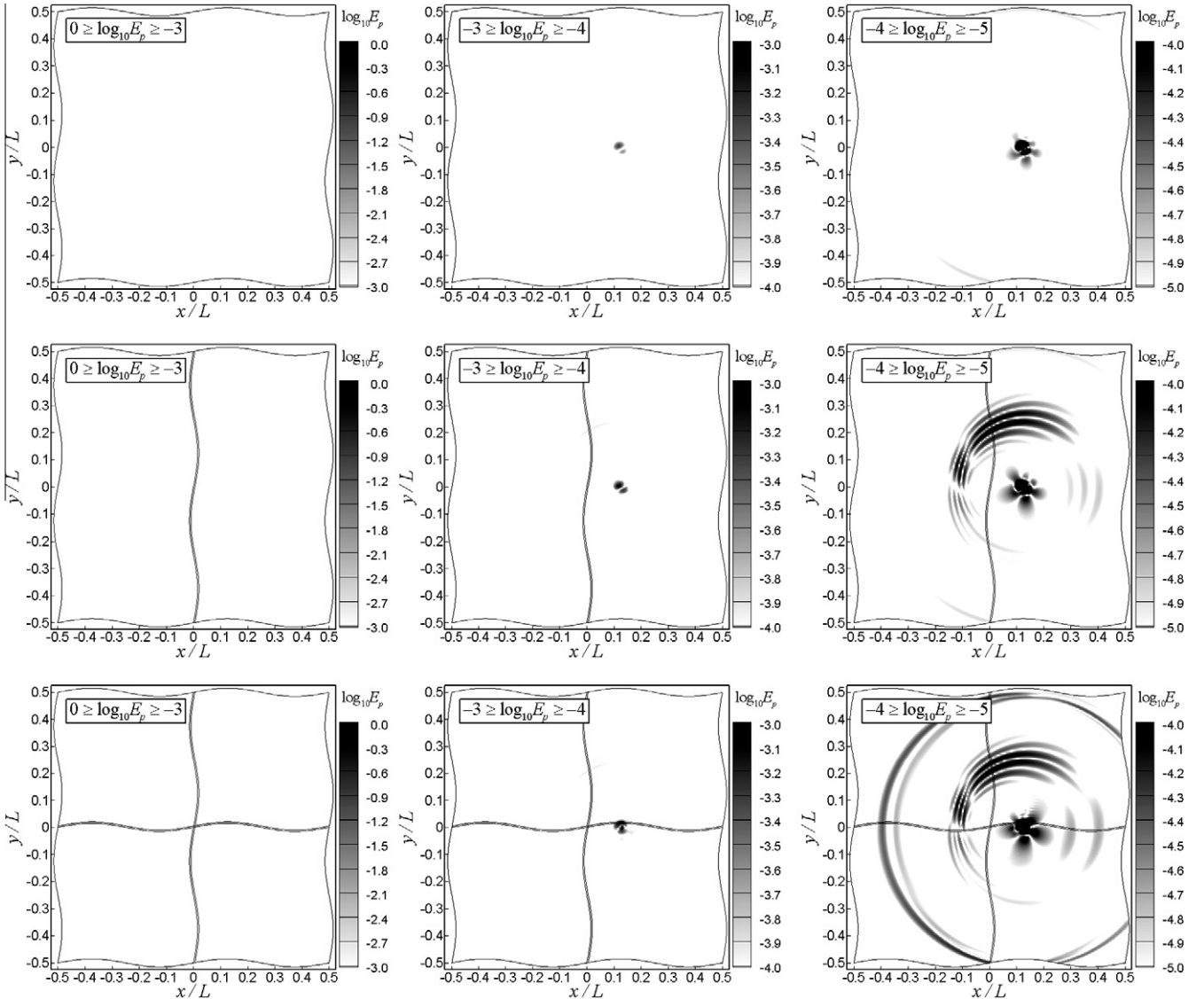


Fig. 10. Contour plots of normalized error (E_p) given by Eq. (6.4) in logarithmic scales at $u_{\infty}t/L = 0.25$. Calculations with $k_1 = 50$ and $k_2 = 0.24$ in Eq. (6.1). Original domain (top) split into two subdomains (middle) and four subdomains (bottom). Curves in the pictures: subdomain boundaries in a curvilinear grid with $\varepsilon = 3$ in Eq. (6.2).

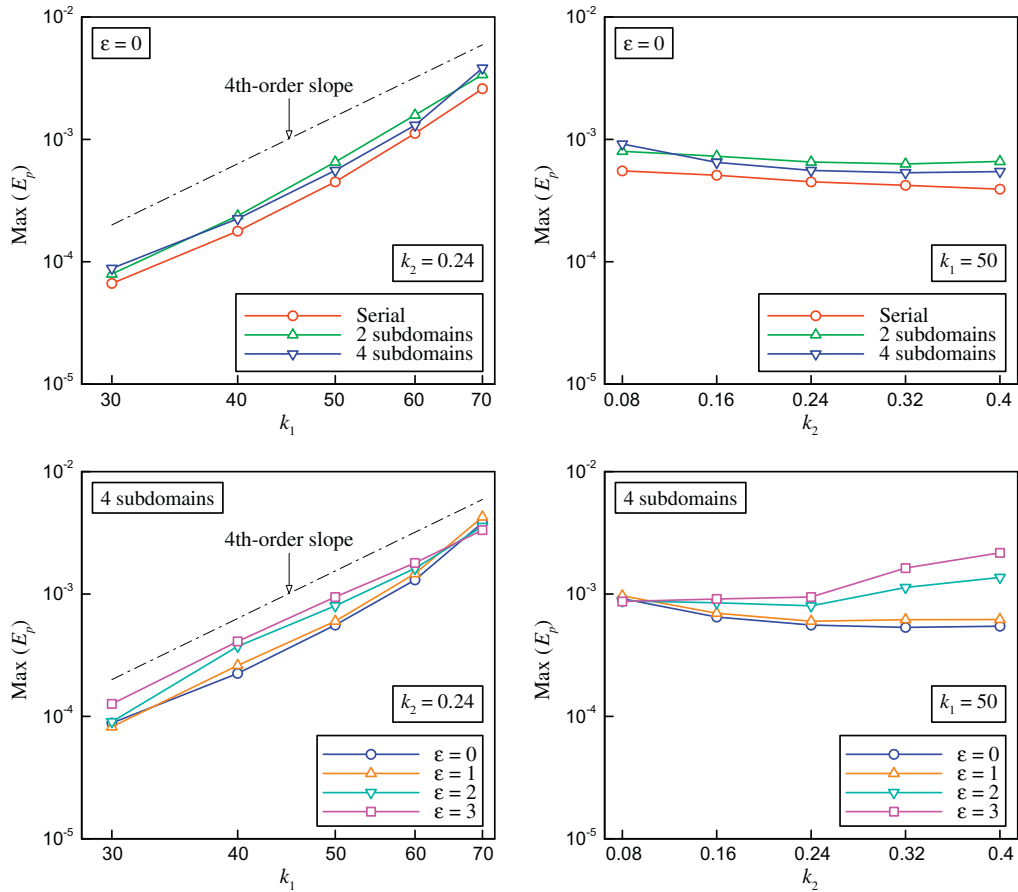


Fig. 11. Plots of the maximum values of normalized error (E_p) given by Eq. (6.4) (obtained at $u_\infty t/L = 0.25$) with different values of k_1 (left) and k_2 (right) in Eq. (6.1). The effects of domain decomposition (top) and grid deformation (bottom).

6.3. Parametric studies

A parametric study on the errors is carried out for different values of the constants k_1 and k_2 on L_∞ -norm basis as shown in Fig. 11. The constant $k_1 = 30$ to 70 corresponds to the diameter of the vortex resolved by 28.7 to 12.3 cells. The other constant $k_2 = 0.08$ to 0.4 changes the minimum pressure at the vortex core from $p_{\min} = 0.9955p_\infty$ (linear) to $p_{\min} = 0.8924p_\infty$ (nonlinear). The top pictures of Fig. 11 demonstrate the error profiles on three different domain topologies, and the bottom ones for four different levels of grid deformation. The left pictures of Fig. 11 confirm the global fourth-order accuracy of the system since k_1 represents the relative size of grid cells to the vortex diameter, i.e. a grid convergence test. Overall it can be seen that the level of errors from the proposed parallel computing approach is consistently within the same order of magnitude with that from a serial calculation across a range of different vortex sizes, strengths and the levels of grid deformation. However, the errors tend to grow noticeably when the vortex becomes nonlinear and the meshes are increasingly deformed, although the deviation is still within one order of magnitude. It should be noted as a comparison that the earlier work in [9] showed a sharp rise in error (based on L_2 -norm rather than L_∞) by two orders of magnitude at the boundary crossing of a linear vortex on a uniform grid.

The aggregate CPU time elapsed per time step per grid point in an Intel Xeon W3520 quad-core CPU at 2.67 GHz is measured and listed in Table 5 for each domain topology. This reveals extra costs mostly for implementing the compact filter in the predictor–corrector mode used in the split domain cases. It is estimated that a parallel calculation in the two-dimensional Euler simulation may

Table 5

Aggregate CPU time elapsed per time step per grid point in two-dimensional isentropic vortex convection.

	Single domain	Split in one direction	Split in two directions
Aggregate CPU time	1.042 μ s	1.136 μ s	1.263 μ s
Relative ratio	1	1.090	1.212

lead up to 21% overhead compared to the corresponding serial calculation. The overhead will grow in three-dimensional cases. However, in Navier–Stokes calculations where the portion of time required for filtering becomes much less, the overhead will be substantially smaller. Inter-node communication normally takes up most of the overhead in three-dimensional Navier–Stokes calculations carried out in parallel across a large number of CPU nodes, which is discussed in the next section.

A test on the performance of the proposed compact filter is carried out to confirm the optimal cut-off wavenumbers $\kappa_C = 0.87\pi$ and $\kappa_C^* = 0.8\pi$ suggested in Section 3. Fig. 12 shows the resulting error distribution varying with κ_C and κ_C^* from the calculation of the two-dimensional vortex convection. This parametric study consisting of 273 cases confirms that the suggested combination of cut-off wavenumbers creates the lowest error.

7. Application to direct numerical simulations

For the simulation of the full Navier–Stokes equations, a multi-block compressible code in generalized cylindrical coordinates is

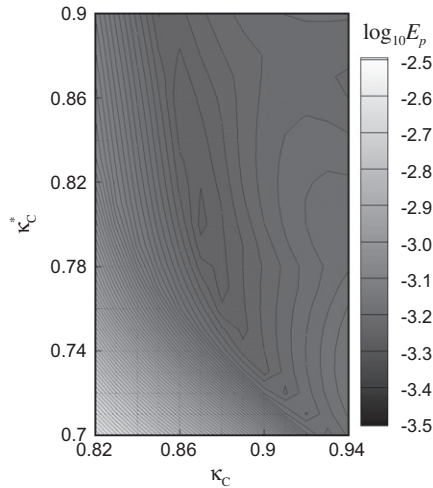


Fig. 12. Variation of normalized error (E_p) given by Eq. (6.4) with different combinations of cut-off wavenumbers (κ_C and κ_C^*). Calculations with $k_1 = 50$ and $k_2 = 0.24$ in Eq. (6.1) with the domain split into four subdomains. $\varepsilon = 0$ in Eq. (6.2).

used, which employs the compact finite difference scheme and filter proposed here for the discretization of the spatial derivatives in the streamwise and lateral directions. In the spanwise, or azimuthal, direction, a pseudo-spectral approach using the FFTW3 libraries is employed, enabling an axis treatment that exploits parity conditions of individual Fourier modes [16]. For the time integration a low-storage fourth-order Runge–Kutta scheme is used [17]. To ensure stability over long run-times, a skew-symmetric splitting of the nonlinear terms is used [18]. Non-reflective characteristic boundary conditions are implemented to allow high-fidelity aeroacoustic simulations to be performed [19]. To take full advantage of massively parallel HPC systems, the code is parallelized using MPI libraries based on a domain decomposition approach in the streamwise and radial directions.

7.1. Tollmien–Schlichting waves

The computation of Tollmien–Schlichting (TS) waves in a flat plate (zero pressure gradient) boundary layer is chosen as a challenging test for direct numerical simulation (DNS) using the proposed parallel compact differencing-filtering system. A viscous hydrodynamic instability is responsible for the growth of small amplitude waves (TS waves) in a boundary layer. Therefore, an accurate representation of the physical mechanisms in the near-wall region is crucial for capturing the correct behavior of the

waves, such as phase and amplitude distributions and, in particular, the growth rates. Moreover, the accurate treatment of the viscous terms is of crucial importance in this particular test case and the suitability of the proposed parallel computing approach for viscosity dominated problems can therefore be assessed.

The simulation of TS waves can be performed on a Cartesian grid in two dimensions. Therefore, the current case uses the two-dimensional form of Eqs. (5.1)–(5.3). The calculation setup follows case 1 from Fasel and Konzelmann [20] with the computational domain spanning $0.2683 \leq x/L \leq 4.5$ and $0 \leq y/L \leq 0.15$, where L is a reference length. For the current investigation, two grids are used: (i) a ‘finer’ grid with 480 and 120 points in the streamwise and wall-normal directions, respectively, resulting in a uniform streamwise spacing of $\Delta x/L = 0.00885532$ and the smallest wall normal spacing at the wall of $\Delta y/L = 3.3e-4$; (ii) a ‘coarser’ grid with 280×88 points with a uniform streamwise spacing of $\Delta x/L = 0.01514893$ and the smallest wall normal spacing at the wall of $\Delta y/L = 5e-4$. The free-stream Mach number was set to 0.25 in order to minimize the effect of compressibility while still allowing for practical time steps. The disturbances are introduced via volume forcing as described in [21]. Three simulations are conducted: (i) using the proposed compact finite difference (FD) scheme and filter (denoted by C_FD-C_F), (ii) using the proposed compact FD scheme and the wavenumber optimized 11-point explicit filter proposed by Bogey and Bailly [22] (with a 50% of weighting in order to avoid excessive damping) after every full Runge–Kutta cycle (C_FD-EXP_F), and (iii) using a standard fourth-order central FD scheme with the same explicit filter as in (ii) (ST_FD-EXP_F). All simulations are conducted using 16 computing cores, decomposing the computational domain in eight streamwise and two wall-normal subdomains.

The current data are compared to the incompressible results from [20]. The amplification rate of the inner maximum of the streamwise velocity component is shown in Fig. 13 as a function of Reynolds number based on displacement thickness, Re_δ^* . It can be noticed that for all results obtained using the compressible DNS code, a streamwise modulation of the amplification rate can be observed, not seen in the incompressible data. Additional simulations of supersonic instability waves did not show this behavior, suggesting that this modulation is most likely due to acoustic reflections from the outflow boundary. However, this phenomenon does not appear to affect the growth-rates of the TS waves and therefore additional studies with outflow sponges or larger streamwise domains were not performed.

On the fine grid, all three simulations produce virtually identical results and agree well with the reference DNS. However, considerable differences between the solutions can be observed on the coarse grid. While the proposed C_FD-C_F still predicts the TS wave

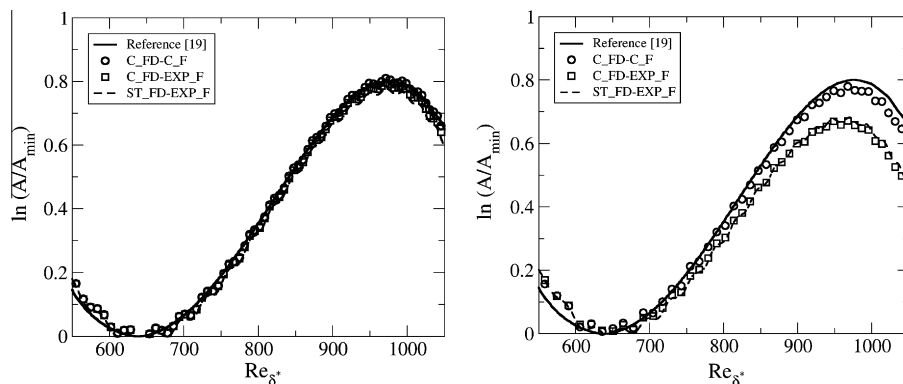


Fig. 13. Amplification rate of the inner maximum of the streamwise velocity component $\ln(A/A_{\min})$ obtained for two-dimensional DNS on fine grid (left) and coarse grid (right).

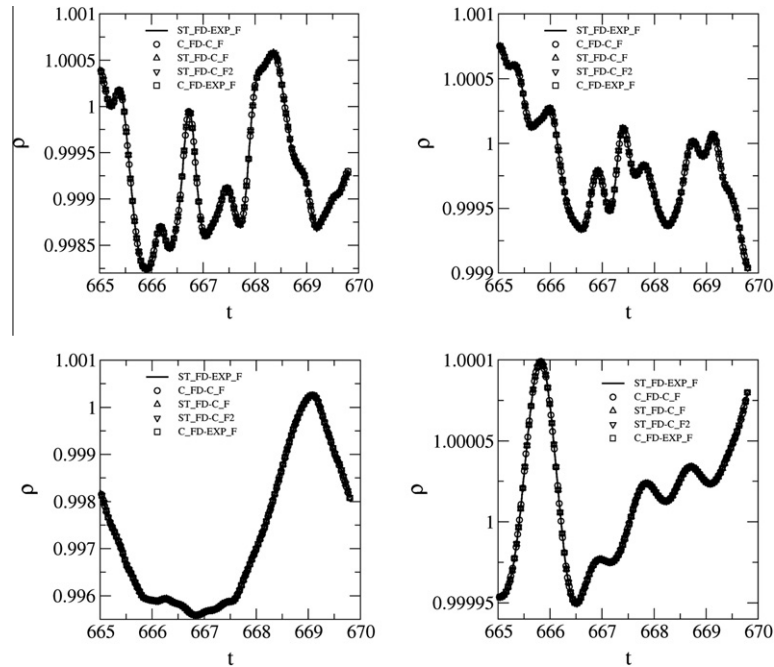


Fig. 14. Time history of density obtained from DNS of a turbulent jet using various combinations of schemes and filters. Locations are $(z,r) = (2R,2R)$ (top left), $(z,r) = (2R,3R)$ (top right), $(z,r) = (12R,2R)$ (bottom left), and $(z,r) = (49.15R,34.41R)$ (bottom right).

growth in good agreement with the reference data, C_FD-EXP_F and ST_FD-EXP_F under-predict the growth rate considerably. This implies that the explicit filter from [22], albeit high-order and wavenumber optimized, is too dissipative for the coarse grid case. Thus, the high wavenumber resolution of the compact scheme combined with the high cut-off wavenumber of the compact filter allows more accurate prediction of the TS wave growth rate on a coarse grid than the standard scheme with the explicit filter does. Overall, it can be concluded from the current test case that the proposed parallel compact scheme and filter are capable of accurately representing the development of hydrodynamic instabilities.

7.2. Direct numerical simulation of a turbulent jet

The final test case is intended to assess the applicability of the proposed parallel computing approach for fully turbulent flows and for direct noise calculations. To that end, a yet unpublished turbulent jet simulation that has recently been conducted using the standard difference scheme and the explicit filter in the current solver is continued with the proposed parallel compact scheme and filter. The ultimate goal of the jet simulation is to perform direct noise calculations, i.e. simulating directly both hydrodynamic and acoustic fields. This requires a large computational domain with sufficient resolution to capture all relevant length scales in the hydrodynamic region and all acoustic wavelengths of interest in the far field.

The computational domain comprises five blocks which can be classified as subdomains containing: flow inside the nozzle (block 1), jet development downstream of the nozzle exit (blocks 2–4), and co-flow and acoustic field upstream of the pipe exit (block 5). For the current study, only blocks 2–4 are of interest. The combined size of these blocks is $110R$ from nozzle exit to outflow, using 2808 grid points with minimum spacing of $0.009R$ at the nozzle exit and maximum spacing of $0.1R$ at the outflow boundary, where R denotes the nozzle radius. The physically useful part of the computational domain in the streamwise direction is up to $79R$, beyond which the zonal characteristic boundary condition [19] (using the last 180 grid points) is activated. In the radial direction,

the computational domain extends from the axis to $80.5R$, using a total of 918 grid points with minimum spacing in the jet shear layer of $0.0026R$, and with the maximum radial spacing chosen such that acoustic waves up to Strouhal number $St_D \sim 2$ (based on the jet velocity and diameter) are resolved with at least 10 grid points. In the azimuthal direction 64 Fourier modes are employed (corresponding to 130 collocation points in physical space), resulting in a total of 408 million grid points. The simulation is conducted using 14,208 cores, dividing the computational domain only in the streamwise and radial directions and thus resulting in subdomains with 13×17 points for each Fourier mode or spanwise plane. The Reynolds number is 7700 based on the nozzle exit velocity and diameter, and the jet Mach number is 0.84.

Four simulations are conducted on the same grid: (i) continuing the original case with the same numerical methods, the standard fourth-order central finite difference (FD) scheme and the 11-point explicit filter [22] (with a 20% of weighting in order to avoid excessive damping) after every full Runge–Kutta cycle (denoted by ST_FD-EXP_F), (ii) using the proposed compact FD scheme and filter (C_FD-C_F), (iii) using the standard FD scheme with the proposed compact filter without altering the cut-off wavenumber at the subdomain boundaries (ST_FD-C_F2), and (iv) using the proposed compact FD scheme with the explicit filter (C_FD-EXP_F). All cases are run for 4.8 non-dimensional time units (based on nozzle radius and nozzle exit velocity) using the same non-dimensional time step of $u_{exit}\Delta t/R = 0.0012$. The initial condition of the current simulations is given by the earlier DNS data obtained at $u_{exit}t/R = 665$.

In Fig. 14, time histories of the axisymmetric mode of density are shown for all simulations at four different locations. All locations mark a characteristic region in the flow, with $(z,r) = (2R,2R)$ being within the initial shear layer of the jet, $(z,r) = (2R,3R)$ being just outside of the initial shear layer of the jet, in the acoustic near field, $(z,r) = (12R,2R)$ being downstream of the potential core within the jet, and $(z,r) = (49.15R,34.41R)$ being in the acoustic far field in the main radiation direction. At all locations, the results obtained from all numerical methods agree with each other for the entire time series, suggesting that neither the hydrodynamic nor

the acoustic field are affected by the choice of scheme or filter. The proposed parallel compact scheme and filter appears to work satisfactorily.

In order to assess whether the acoustic field has experienced any degradation by applying the proposed parallel computing approach, instantaneous contours of the streamwise density gradient are shown in Fig. 15 of the entire computational domain for the planes with $\theta = 0^\circ$ and $\theta = 180^\circ$. The contour levels are chosen very aggressively in order to reveal the acoustic waves radiating to the far field. Firstly, it can be observed that all simulations succeed in capturing the acoustic waves without visible reflections from the outer boundaries due to the proper use of non-reflecting boundary conditions [14,19]. The sudden change in the contours downstream of the dashed line is attributed to the use of zonal boundary conditions [19] across which the jet flow is artificially diffused and modified in order to prevent sound waves from reflecting from the exit boundary. Secondly, it is virtually impossible to detect any difference between the combinations of methods used, in particular when comparing the compact scheme with the standard scheme. This is highly encouraging, given that the computational domain is decomposed into 14,208 subdomains with each domain boundary potentially generating an acoustic reflection.

Fig. 16 shows instantaneous contours of the same variable and for the same planes as Fig. 15, although only for the region with $10 \leq z/R \leq 21$ and $0 \leq r/R \leq 5$ in order to zoom into the hydrodynamic field. Larger bounds are chosen for the contour levels in order to highlight the flow structure. Similar to the figures depicting the entire domain, the hydrodynamic field obtained from all simulations is highly comparable. Overall, based on visual inspection of contour plots, the proposed parallel compact scheme and filter appears suitable also for fully turbulent flows. However, it should be noted that some level of acoustic reflections from the subdomain boundaries was observed when scrutinizing the difference between the reference DNS (ST_FD-EXP_F) and all other cases over a longer period of time. Due to inherent differences in the truncation error of each method that will accumulate over time, the flow fields obtained from different methods cannot be identical. Nevertheless, a slightly larger difference between the result from the C_FD-C_F and the reference DNS data does indicate some artifact reflections from the subdomain boundaries. The observed weak boundary reflections occur very quickly after switching to the compact approach. The reflections propagate with the speed of sound and will therefore be seen a few time steps into the simulation using the compact approach. It is speculated at this point that the boundary reflections in the Navier–Stokes cases might be at a

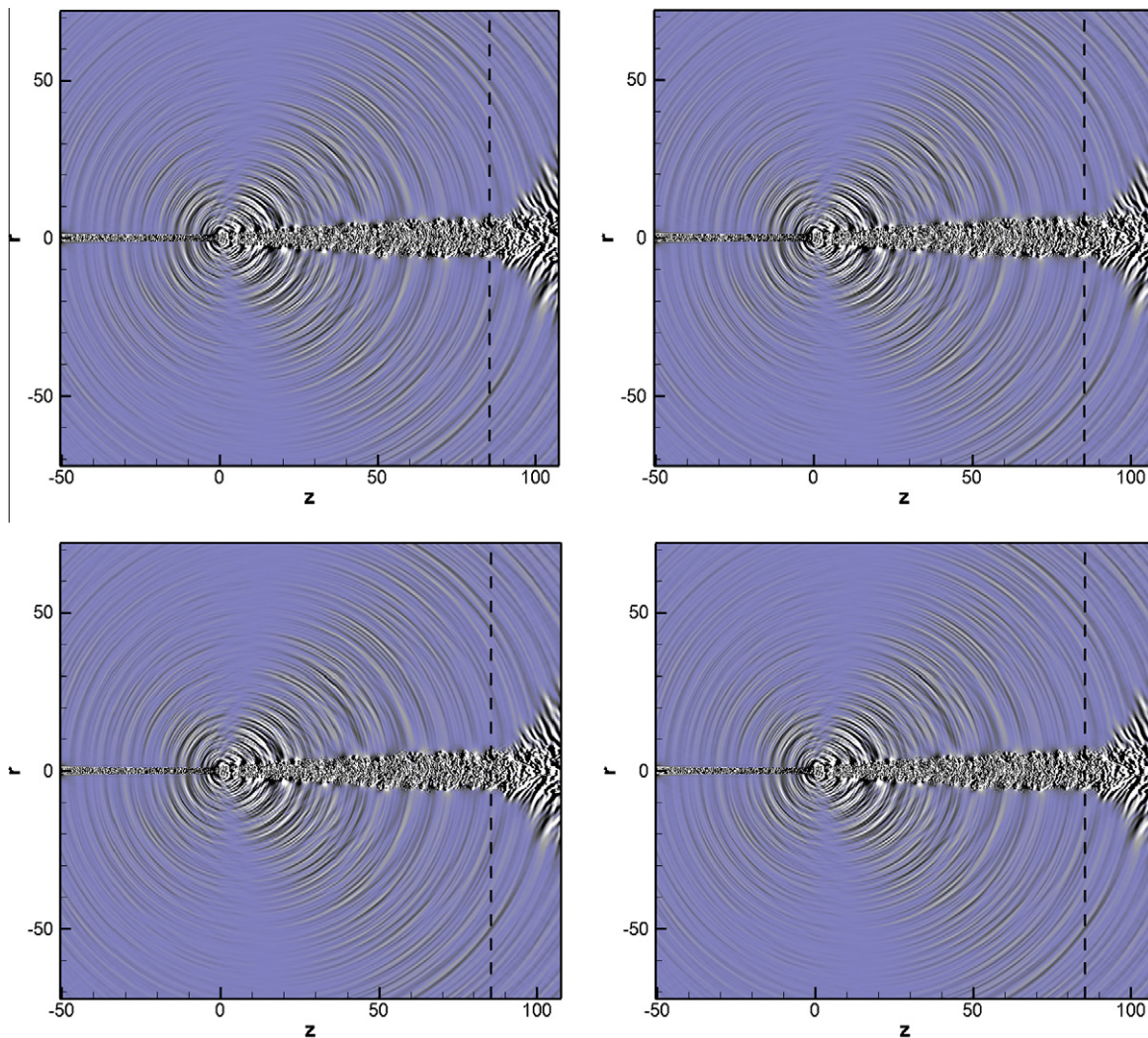


Fig. 15. Instantaneous contours of the streamwise density gradient $\partial(\rho/\rho_\infty)/\partial(z/R)$ with continuous levels from $-5e-4$ to $5e-4$ displaying planes with $\theta = 0^\circ$ and $\theta = 180^\circ$ for entire computational domain: ST_FD-EXP_F (top left), C_FD-C_F (top right), ST_FD-C_F2 (bottom left), and C_FD-EXP_F (bottom right). Dashed line denotes the onset of the zonal characteristic boundary condition [19].

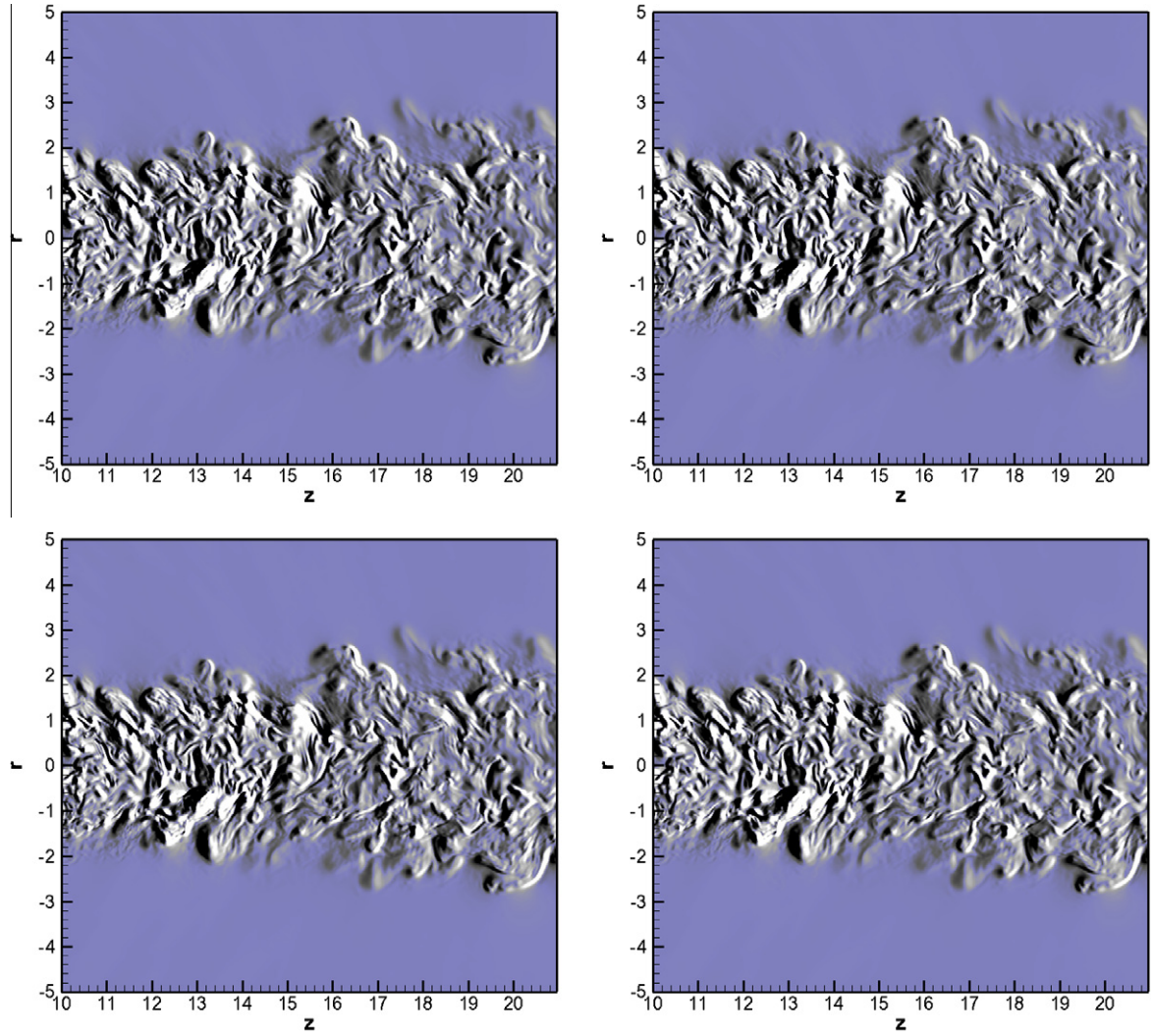


Fig. 16. Instantaneous contours of the streamwise density gradient $\partial(\rho/\rho_\infty)/\partial(z/R)$ with continuous levels from -0.1 to 0.1 displaying planes with $\theta = 0^\circ$ and $\theta = 180^\circ$ for $10 \leq z/R \leq 21$ and $0 \leq r/R \leq 5$: ST_FD-EXP_F (top left), C_FD-C_F (top right), ST_FD-C_F2 (bottom left), and C_FD-EXP_F (bottom right).

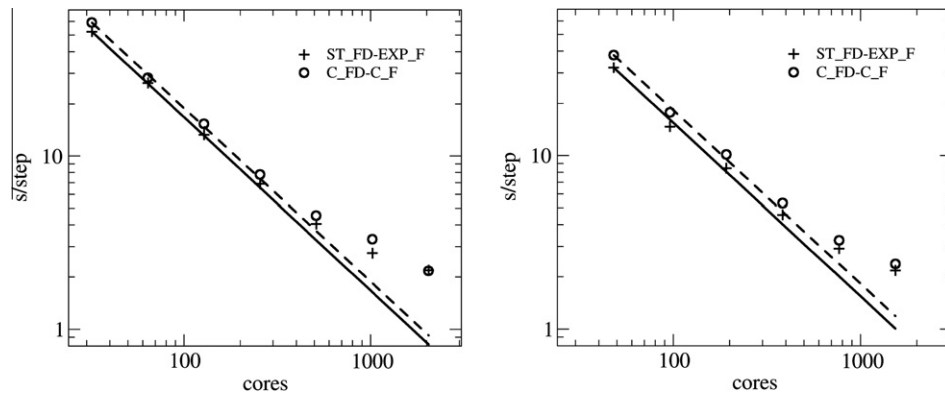


Fig. 17. Performance study in CRAY XT4 (left) and CRAY XT6 (right) systems using the proposed parallel compact scheme and filter (C_FD-C_F), and a standard scheme with an explicit filter (ST_FD-EXP_F). The lines denote linear scaling for respective cases.

higher level than those in the Euler cases discussed in Section 6 since the Navier–Stokes cases involve second derivatives in the viscous terms, which could have contributed to the boundary

reflections. This potential issue is not explicitly taken into account in this paper but a further investigation on the effects of viscous terms would be beneficial.

8. Performance study on distributed memory system

The performance of the DNS code using the proposed parallel compact scheme and filter compared to the standard scheme and explicit filter is tested on two different configurations (CRAY XT4 and XT6) of the UK national supercomputing facility HECToR. The CRAY XT4 uses quad-core AMD Opterons, while the XT6 has 24 computing cores per node. A reasonably large test case, typical of production runs, is set up with 1600 and 800 points in the streamwise and lateral directions, respectively, and 32 Fourier modes for the spanwise direction, resulting in a total of 84.5 million collocation points. For this case, the simulations could not be run on less than 32 cores on the XT4 and 48 cores on the XT6 due to memory requirements.

In Fig. 17, the run-time in seconds per time step, i.e. for a full five-step Runge–Kutta cycle, is shown for core numbers between 32 and 2048, and 48 and 1536 on the XT4 and the XT6, respectively. On the quad-core system, it can be seen that the proposed parallel compact scheme and filter (C_FD-C_F) requires approximately 12% more time per time step than the standard scheme and explicit filter (ST_FD-EXP_F) when using 32 computing cores. This is more than acceptable considering that due to the significantly better wavenumber resolution characteristics of the compact scheme and filter a coarser grid could be used. The small difference (12%) in computational cost is most likely attributed to the bandwidth-limited computer used (HECToR) and, thus, higher algorithmic cost does not cause a considerable increase in overall cost which includes communication between processes. As the number of computing cores is increased, good scaling is found for the code regardless of the spatial discretization scheme up to 1024 cores, beyond which a larger case would need to be considered for efficient parallelization. Encouragingly, for the highest core number tested the difference in run-time per time step between both numerical methods tested is vanishingly small, implying that the parallel efficiency of the proposed C_FD-C_F is at least as good as that of the ST_FD-EXP_F.

The test on the XT6 is performed in order to assess the parallel performance of the proposed method on the latest ‘many-core’ architecture in light of these systems becoming ubiquitous. The right picture of Fig. 17 shows that the scaling of the proposed C_FD-C_F is also excellent on a system with many cores per node and very competitive with the scaling of ST_FD-EXP_F. The superior speed-up of the C_FD-C_F at higher core numbers can be explained by their higher algorithmic cost compared to the ST_FD-EXP_F. Codes with higher ratios of algorithmic operations over communication fare better on current bandwidth-limited computing systems. It can therefore be concluded that the parallel compact scheme and filter scales efficiently on current high performance computing systems, approaching the computational cost of standard schemes with explicit filters at higher core counts due to a better FLOPs/communication ratio.

9. Concluding remarks

The proposed fourth-order compact finite difference scheme based on three halo points combined with a newly developed sixth-order compact filter is successfully implemented in massively parallel computations. The error levels of the proposed parallel calculations are shown to be comparable to those of equivalent serial single-domain calculations and/or those of conventional parallel calculations using standard methods. This is a notable improvement compared to one of the latest publications that resulted in a sharp rise of parallelization induced errors by two orders of magnitude (see [9]). The pentadiagonal matrix systems of the proposed approach including the predictor–corrector

steps in the filtering process incur only a moderate increase in run-time per time step in fully three-dimensional Navier–Stokes calculations compared to the standard methods without a matrix inversion process involved. The higher resolution of the proposed compact scheme and filter implies the use of substantially fewer grid cells than the standard methods use, which is one of the major benefits of employing the proposed approach. It is also demonstrated that the speed-up rate of the proposed approach in a massively parallel computing environment is equivalent to or even better at 1000+ cores than that of the standard methods.

However, it is learned during the course of tests that the proposed parallel compact scheme still generates artifact acoustic reflections when strong vortices resolved by a small number of grid points pass through subdomain boundaries. Although the extra errors in Euler cases are significantly reduced to a level smaller than the original errors from serial calculations, slightly more substantial reflections are observed in the fully turbulent Navier–Stokes cases. It is conjectured that the increased amplitude of reflections might be attributed to the viscous terms containing second derivatives which go through the process of numerical differentiation (applying the proposed compact scheme) twice. Therefore, there is a possibility of additional reflections due to the viscous terms despite their dissipative nature. This leaves the authors with scope for further work on parallel compact schemes, particularly when applied to subsonic jet noise which is more sensitive to accurate representation of phase and amplitude characteristics of vortices than others. Improving the uniformity of resolution characteristics of the compact scheme across subdomain boundaries will be necessary to further reduce the boundary reflections to a level that is convincingly negligible in the future work.

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